

Is human capital the sixth factor?

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Abstract

A six factor model designed for capturing the size, value, profitability, investment and human capital patterns in average portfolio returns performs better than both [Fama-French \(1993\)](#) three and [Fama-French \(2015\)](#) five factor model, and performance of the model is not sensitive to the technique its factor variables are defined. The six factor model's main problem is its failure in capturing the average returns on “microcap with low value stocks that are highly profitable invest aggressively for asset growth but invest much lesser for human growth” and “microcap with unprofitable stocks whose returns behave like those low value firms with conservative investment”. Study finds the investment factor (CMA) of Fama-French (2015) five factor model as the redundant factor for describing the portfolio average returns in the study sample.

Is human capital the sixth factor?

Introduction

Today global investors' investment focus is not fully concentrated into the developed markets rather parts large share in the emerging markets. In the recent past, the emerging markets show evidences of significant development and growth¹, which draw the attention of the several investors and researches globally to access the risk inherent in these markets. Rapid development and growth of the emerging markets give birth to various unidentified factors in the capital markets which are different from developed markets. That gives a tough challenge to the investors and researchers to identify such unidentified factors prevailing in the emerging capital markets and makes asset pricing difficult. Over the years, several stock market anomalies have emerged and they also pose a tremendous challenge to the asset pricing framework. Emergence of new anomalies overturns the CAPM's success and other multifactor models in capturing the risk return relationship. Even Fama-French (2015) is unable to capture the risk return relationship for the micro stocks or smallest portfolio. Likewise most of the studies (Hou, Xue, Zhang (2014); Clarke (2016); Chiah, Chai, Zhong & Li (2016), Fama-French (2016) etc.) carried out across the world find Fama-French (2015) five factor model is not able to capture all the risk return variations that justify the scope for the new more robust models. Most of the prominent asset pricing models whether it is CAPM or Fama-French factor models have ignored human capital in the model. Human capital is an important factor as it represents the wealth of the company and Campbell (1996) introduces human factor as important component of wealth. The study also argues that labour (human capital) captures approximately two third of the GNP and rest is related to the financial wealth. Campbell (1992, 1996) calculates the total wealth by using the equation (1).

$$W_{t+1} = R_{m,t+1} (W_t - C_t) \quad (1)$$

Where W_{t+1} represents the future total wealth that includes human capital, savings & other returns on wealth, the model assumes representative agent economy. The right hand side of the equation (1) shows the return on the investments. Which is calculated by removing the consumption (C_t) component from the total aggregate wealth and then it is multiplied by the return on the aggregate wealth ($R_{m,t+1}$). To linearize the non-linear budget constraint equation

¹Blackrock Investment Institute report(2011) on "Are Emerging Markets the Next Developed Markets?". It is mentioned that the developed market share is declining specially emerging Asian market are booming. The article can be accessed from <https://www.blackrock.com/corporate/en-nl/literature/whitepaper/are-emerging-markets-the-next-developed-markets-us.pdf>

(1) divided by W_t . Then equation (1) is transformed into log transformation to apply first order Taylor approximation (See [Campbell \(1992\)](#)) to estimate the present value of the future returns on wealth. The adjustment between the present and future value for the consumption is shown below in the equation (2).

$$C_{t+1} - E_t C_{t+1} = (E_{t+1} - E_t) \sum_{j=0}^{\infty} \sigma^j_{m,t+1+j} - (E_{t+1} - E_t) \sum_{j=0}^{\infty} \sigma^j_{g,Ct+1+j} \quad (2)$$

Then to solve the Euler equation with budget constraint [Campbell \(1992\)](#) uses [Epstein and Zin \(1989, 1991\)](#) objective function and the resultant excess return of the asset shown below in equation (3).

$$E_t r_{i,t+1} - r_{f,t+1} + V_{ii}/2 = \Theta^* (V_{ic}/g) + (1-\Theta) * V_{im} \quad (3)$$

Where $E_t r_{i,t+1} - r_{f,t+1}$ is the excess return of the asset; V_{ic}/g covariance of consumption divided by the inter temporal elasticity of consumption; V_{im} covariance of market return; Θ & $(1-\Theta)$ represents the weight and $V_{ii}/2$ represents the weighted average of the two covariance's (V_{ic}/g & V_{im}). Then the cross sectional asset pricing formula without consumption derived by combining the log linear Euler's equation and budget constraint equation (See [Campbell \(1996\)](#)) and the human capital introduced in the model as shown below in equation (4).

$$R_{m,t+1} = (1 - V_t) * R_{a,t+1} + V_t * R_{y,t+1} \quad (4)$$

V_t represent the human wealth to the total wealth ratio; a financial asset²; y human wealth³; $R_{a,t+1}$ returns on financial asset and $R_{y,t+1}$ returns on human asset. This model proposes to bring the human capital to tests the role of human asset along with financial asset in determining the returns and it is tested using the equation (3) risk premium formula as shown below in equation (5) below.

$$E_t r_{i,t+1} - r_{f,t+1} + V_{ii}/2 = \Theta^*(1-v)*V_{ia} + \Theta_v * V_{iy} + \{ \Theta^*(1-v) - 1 \} * V_{ih} \quad (5)$$

Where,

$$V_{iy} = \text{Cov} [r_{i,t+1} - E_t r_{i,t+1}, (E_{t+1} - E_t) \sum_{j=0}^{\infty} \sigma^j_{gy,t+1+j}] \quad (6)$$

²Financial assets are the non-physical and intangible assets like stocks, bonds and bank deposits that derive their values from the contractual claims of what they represent. They are more liquid than tangible assets and can be traded in the capital markets.

³Human wealth represents the intangibles of the company that are incorporated in the financial statements of the company but it decides the wealth or value of the company in term of measuring the economic value of the company's employee skill sets etc.

$\Theta^*(1-v)$ and Θ_v weights for market return and human capital respectively and model will fail when v is zero. The study results indicate that omission of the human factor may lead to serious problems. Other studies by [Fama and Schwert \(1977\)](#), [Jagannathan and Wang \(1996\)](#), [Jagannathan et al. \(1998\)](#), [Rosett \(2001\)](#) and [Qin \(2002\)](#) use human capital to explain the cross sectional variations of the security returns. The present study focuses on one of the major emerging Asian economies – India⁴ to examine the role of human capital⁵ in asset prices. In this study various existing multifactor asset pricing models like [Fama-French \(1993\)](#) three factor and [Fama-French \(2015\)](#) five factor model are experimented in a time series setup to test the risk return relationship. Then the results are compared with our newly developed six factor asset pricing model with human capital to justify the superiority of our model. The present study is novel for several reasons: First, it is six factor descriptions; second, no comprehensive asset pricing study is done with human capital in Asian emerging markets especially in India. Perhaps this is the first study to examine whether portfolio returns are affected by the human capital in Indian context. Third our study period and methodology used are completely different from the previous studies.

⁴ Among emerging nations, India is very important and unique due to several reasons. Several Institutional and Non institutional differences exist between India and developed markets in terms of settlement cycle, trading timings and mostly Indian markets are order driven market. Only 1% of the Indian household savings are invested in the capital market and major portion is invested in intraday. Indian capital market has the potential to become the future developed market (See Blackrock investment institute report (2011)). Due to globalization now Indian market is integrated to the other major global markets (See Narayan et al. (2014), Ahmad et al. (2013), Lim and Kim (2011), Awokuse et al.(2009), and Tiwari and Kyophilavong (2014)). Several factor models that are successful in developed countries become failure in Indian context or the vice versa for example FTF model is rejected by GRS test by Fama-French (2012, 2015) in US context whereas FTF model is accepted by GRS test by Connor & Sehgal (2003) in Indian context . Further study by Griffin (2002), Fama and French (2012, 2015c) findings suggest that Fama-French factor models yield different model for regional and country level due to influence of several other unknown factors. That fully justifies the need of further comprehensive studies in the Indian market.

⁵ Company's balance sheet generally excludes the intangible assets like Human capital of the company and human capital factor of the company add much more reliable information about the value of the company that helps the investor to take rational decision. Hence, some investor will take advantage in the market those have information about the human capital of the company than the rest of the market. Nowadays it is also true those companies are building competitive advantages by investing in the intangible assets like human capital and knowledge base within the company rather than investing in the physical assets. Further study by Mörling, A., & Larsson, D. (2015) find positive causal relationship between the human capital and stock returns. Second report published by world economic forum on human capital (2016) ranks India far below in HCI (Human Capital Index) compared to other emerging markets. Further the report also suggests that human capital of India lacks innovations and talents. This suggests that human capital in India context is very important and investors can gain abnormal returns by mapping their investment strategy to the human capital of the company.

Literature review

The single factor model CAPM ([Sharpe \(1964\)](#), [Lintner \(1965\)](#) & [Mossin \(1966\)](#)) is a notable breakthrough in the literature of asset pricing. The CAPM argues that risk can be classified into two group's namely systematic risk (market risk or beta) and unsystematic risk⁶. Systematic risk which cannot be diversified and other category can be diversified; hence systematic risk should be considered in calculating the expected portfolio returns. However capital markets are more complete in operation and with single factor beta it is quite impossible to capture all the risk return variations. Hence, CAPM was criticised and challenged by [Fisher Black \(1972\)](#), [Fama and MacBeth \(1973\)](#), [Ross \(1976\)](#), [Banz \(1981\)](#), [Reinganum \(1981\)](#), [Gibbons \(1982\)](#), [Basu \(1983\)](#), [Shanken \(1985\)](#) and [Bhandari \(1988\)](#). Further number of sizeable studies by [Bhandari \(1988\)](#), [Fama-French \(1988\)](#), [Gibbon, Ross and Jay Shanken \(1989\)](#), [Jaffe, Keim & Westerfield \(1989\)](#), and [Lo & MacKinley \(1990\)](#) etc., The above studies find that size of the company is an another important variable along with beta in explaining the expected portfolio returns. Similar type of study by [Basu \(1983\)](#), [Reinganum \(1981\)](#) etc., finds P/E ratio and [Bhandari \(1988\)](#), [Narayanaswamy and Phillips \(1987\)](#) etc., find leverage are the other important factors that explain portfolio expected return variations. Then [Fama-French \(1993\)](#) introduce more stable three factor model from the previous studies with beta, size, and value. In Indian context notable studies by [Connor and Sehgal \(2003\)](#), [Sehgal and Balakrishnan A \(2013\)](#) studies find that three factor model is better than CAPM in capturing the risk return relationship. Then [Fama-French \(1993\)](#) three factor model is criticised and challenged by several studies by [Haugen and Baker \(1996\)](#); [Cohen, Gompers, and Vuolteenaho \(2002\)](#); [Fairfield, Wiseman, and Yohn \(2003\)](#); [Titman, Wei and Xie \(2004\)](#); [Novy-Marx \(2013\)](#), [Hou, Xue, and Zhang \(2014\)](#) etc., due to the several other unidentified capital market anomalies. Finally [Fama-French \(2015\)](#) introduce five factor model with beta, size, value, profitability and investment factor that are able to capture more risk return variations than the [Fama-French \(1993\)](#) three factor model. [Fama-French \(2015\)](#) five factor model is unable to capture the risk return variations for the micro stock portfolios. Likewise most of the studies ([Hou, Xue, Zhang \(2014\)](#); [Clarke \(2016\)](#); [Chiah, Chai, Zhong & Li \(2016\)](#)) done across the world find [Fama-French \(2015\)](#) five factor model is not able to capture all the risk return variations that justify the scope for the new and more robust model.

⁶It is the category of risk that a particular company or industry that possess and can be reduced by diversification.

All of the notable studies mentioned above have not considered human capital factor in the model. Human capital factor is considered to be a key factor and explains the wealth of the company. It is very difficult to measure the human capital factor very accurately. [Campbell \(1996\)](#) study introduces the human capital factor and argues that omission of the human capital from the model may lead to problem in the asset pricing as explained in the introduction section. [Jagannathan and Wang \(1996\)](#) include the human capital return instead of the market portfolio return and find that the unconditional model (CAPM) is able to explain over 50 % of variations in the cross section returns. [Jagannathan et.al. \(1998\)](#) find that labour income is an important variable to define aggregate risk which is not captured by the traditional stock return index factor or market returns. Interesting part of the study is that labour beta drives out size factor but not the value factor. [Rosett \(2000\)](#) finds economically and statistically human capital (labour stock) has a direct positive relation with the equity investment risk. The study uses three standard models, both market and accounting based risk measures variables to arrive at the said conclusion. [Qin \(2002\)](#) addresses why the asset's market beta is unable to capture risk associated with the human capital. The study derives an intertemporal model that relates market beta and human capital beta linearly with the asset returns. [Pantzalis& Park \(2009\); Edmans \(2011\)](#) find that human capital is related to the stock returns but market cannot correctly price the human capital for the small size stock due to small firm effect⁷. In Indian context [Shijin et.al \(2010\)](#) find causal relationship between the asset returns and labour income, the study is mainly based on the nifty 50 index. The study finds that human capital is a significant factor that can affect the asset prices and adds the human capital factor to the model can predict more variations. Though the study is done for only market index and it does not consider different stock portfolio returns. From the above rigorous literature review it is concluded that human capital does affect the asset prices and in Indian context, a comprehensive study on human capital and stock returns is needed and none of the previous studies address whether portfolio returns are affected by the human capital or not in Indian context. So, the present study will bridge the existing gap by developing a new model with human capital using different study period and methodology from the previous studies that can capture the risk return variation better than the existing asset pricing models.

⁷A theory that holds a company with smaller market capitalization outperforms big market capitalization companies.

Data and Methodology

Data

The study uses monthly data for the total of 431 companies out of 500 listed on BSE-500 companies that are taken from the Bloomberg database for the period from July, 2004 to November, 2016. Market capitalisation (MC) is used as the proxy for the size factor variable and P/B ratio⁸ the inverse of the BE/ME ratio is employed as the proxy for the value factor variable. Total asset growth (TA) is the proxy for Investment & it is calculated by the formula $[(TA_t - TA_{t-1}) / TA_{t-1}]$ as similar to Hou, Xue and Zhang (2015) and for profitability, return on equity (ROE) is used as the proxy which is similar to Haugen and Baker (1996). Studies by Hansson (2004), Draca, Machin & Reneen (2011), Bell & Machin (2016) use average wages for the proxy of human, however our study uses the growth of salary & wages as proxy of human capital as a measure of total investment made by company for human development as using growth of wages and salaries gives more clear picture about how much company compensates more or less on human asset. Sixth factor variable is human capital (HC) growth factor variable that acts as the proxy for human capital (salary and wages growth) of the company and it is calculated by the formula $[(HC_t - HC_{t-1}) / HC_{t-1}]$. The study uses 91 days T-Bills as the risk free rate (R_f) and BSE200 returns as the proxy for market return (R_m).

Portfolio Construction

In this study, the portfolios are constructed using single and double sorting techniques. In the month of June year (t) we form five equally weighted portfolios based on MC by using single sort. Then portfolios are named as M1 to M5. M1 portfolio consists of the bottom 20 % of stocks of small size whereas M5 portfolio consists of the top 20% of stocks of big size stocks. Rest of the intervening portfolios namely M2, M3, and M4 are ranked in between M1 & M5 and each intervening portfolio consists of 20 % of stocks ranked in ascending order of stock size. Then in the month of June⁹ year (t+1) the ranking is revised using the same process and continued every year till 2016. Next, using the same portfolio construction procedures, five equally weighted portfolios are formed based on other variables such as P/B, TA, ROE, and HC.

⁸P/B ratio is the inverse of the BE/ME, the interpretation of the value factor will be exactly the mirror image of the Fama-French (1993, 2015) results.

⁹We give three months lag period, generally in India the financial year ends in the month of March of every year. The lag of three month is more than enough to publish the accounting data and moreover mostly now a days it is digitally made available to all the stakeholders well before.

In the month of June year (t) using the double sorting technique, 25 portfolios are constructed from the intersection of the five MC and five P/B sorted portfolios (obtained from the single sort). Then portfolios are named from P1 to P25. P1 is the portfolio one consists of small size and low value stocks while P25 is the twenty fifth portfolio that contains the stocks of big size and high value. Then each portfolio's monthly mean excess returns are calculated from July, 2004 (t) to June, 2005 (t+1). Next revision of portfolio formation is done in year 2005 and the process of portfolio revision continues till 2016. Finally, mean excess returns on each portfolio is calculated for 149 months from July, 2004 to November, 2016. Similar process is adopted in construction of the other portfolios from the intersection of MC – TA, MC- ROE, and MC - HC respectively. Finally the study has total of 100 portfolios (Dependent variables).

Next, we form portfolios using double sorting technique. In the month of June year (t) sample stocks are ranked on MC and two equally¹⁰ weighted portfolios are constructed namely small and big. Small (S) portfolio consists of bottom 50% of sample the stocks and they are small stocks based on MC while Big (B) portfolio consists of top 50% of the sample stocks and these are big stocks on MC. Next, in the month of June year (t+1) the ranking of portfolios is revised using the same process and continued every year till 2016. Then in the month of June year (t), we again form mimicking portfolios using double sorting technique based on P/B and three value weighted (30:40:30, Fama-French (1993) breakpoints) portfolios are constructed namely low, neutral, and high. Low (L) portfolio consists of bottom 30 % of the sample stocks; whereas High (H) portfolio contains the top 30% of the sample stocks. Intervening 40 % of the stocks are bracketed in neutral (N) group. Then portfolio revision is done in the month of June year (t+1) using the same procedures. The portfolio revision is carried out for whole study period. Next, portfolios are formed on ROE using the same breakpoints as that of P/B classification. Portfolios based on ROE are named as weak, neutral, and robust. Weak (W) portfolio is the one that consists of bottom 30 % of the sample stocks and these stocks (firms) are with low profitability whereas robust (R) portfolio contains the top 30% of sample stocks and these stocks (firms) inherently have high profitability. The portfolio which is in between weak and robust is called neutral (N) and the

¹⁰Fama-French (2015) use MC breakpoints of 10:90 as US stocks are more dominated by the big size stocks but in India most of the prominent studies are done using 50:50 break points (see Connor and Sehgal (2003) , Sehgal and Balakrishnan A (2013), Balakrishnan A & Maiti (2015)). Further studies by L'Her, J. F., Masmoudi, T., & Suret, J. M. (2004); Hou, K., Xue, C., & Zhang, L. (2014); Durand, R. B., Limkriangkrai, M., & Chai, D. (2015) etc., has used 50:50 breakpoints for size in their studies in international context. We use similar 50:50 breakpoint for the MC or size in constructing our study portfolios.

same comprises of 40% of the sample stocks falling between bottom 30% and top 30% of ranking and these stocks (firms) make plausible profitability. Next, three portfolios are formed on asset growth namely conservative (C), neutral (N), and aggressive (A). Conservative (C) portfolio consists of the bottom 30 % of the stocks with low investment stocks whereas Aggressive (A) portfolio contains the top 30% of the aggressive stocks with high investment stocks and intervening 40 % of the sample stocks are with neutral investment stocks and they are in the Neutral (N) portfolio. Finally, we form three portfolios on human capital¹¹. The portfolios are of low (LC), benchmark (BC), and high (HC). Low portfolio consists of bottom 30 % of low compensation stocks with low growth rate on (salary & wages) stocks whereas High (HC) portfolio consists of the top 30% of high compensation stocks with high growth rate on (salary & wages) stocks and rest 40 % of the stocks are named as balanced compensation with balanced growth rate on (salary & wages) stocks placed in the Benchmark (BC) portfolio.

In the month of June year (t) using double sorting technique 6 portfolios are constructed from the intersection of two MC and three P/B sorted portfolios (obtained from the single sort). Then six portfolios are named as S/L, S/N, S/G, B/L, B/N and B/G. S/L portfolio consists of ‘small size & low value’ stocks and B/G portfolio consists of ‘big size & high value’ stocks. Then each portfolio’s value weighted excess returns are calculated from July, 2004 (t) to June, 2005 (t+1). Next revision of portfolio formation is done in year 2005 and the process of portfolio revision continues till 2016. Finally, mean excess returns on each portfolio are calculated for 149 months from July, 2004 to November, 2016. Similar process is followed for the construction of the other portfolios from the intersection of MC- TA, MA- ROE, and MC- HC respectively. Six mimicking portfolios based on MC- ROE cross are named as S/W, S/N, S/R, B/W, B/N and B/R. S/W portfolio consists of ‘small size & low profitability’ stocks and B/G portfolio consists of ‘big size & high profitability’ stocks.

¹¹ In this study, we use salaries and wages as the proxy of human capital. Generally, salaries and wages are fixed by the firms in three scales. They are: (1) low compensation (2) Benchmark compensation (3) Higher compensation. Low compensation is the salaries and wages being fixed below the average salary paid by the industry. On the other hand, if salaries and wages are fixed above the average salary being paid by the respective industry, this is called as high compensation. If salaries are paid in line with industry, that is called benchmark compensation.

Then six mimicking portfolios based on MC-TA are named as S/C, S/N, S/A, B/C, B/N and B/A. S/C portfolio consists of small size and low investment stocks and B/A portfolio comprises of big size and high investment stocks. Finally six mimicking portfolios based on MC- HC cross are named as S/Lc, S/B, S/Hc, B/Lc, B/B and B/Hc. S/Lc portfolio consists of small size and low compensation stocks while B/Hc portfolio consists of big size and high compensation stocks.

Mimicking portfolios

The study uses five mimicking portfolios SMB(Size), LMH(value), RMW(profitability), CMA(Investment) and LcMHc (human capital) that are calculated as explained below:

SMB

SMB stands for small minus big and it mimics the risk with regard to company size. Further, it is calculated as shown below:

$$SMB = (S/L + S/M + S/H)/3 - (B/L + B/M + B/H)/3 \quad (7)$$

LMH

LMH¹² stands for low minus high and it mimics the risk associated with company value. It is computed as under:

$$LMH = (S/L + B/L)/2 - (S/H + B/H)/2 \quad (8)$$

In this study, we use P/B ratio as the proxy of company value instead of BE/ME ratio, which is the inverse of P/B. Hence, interpretation of the results will be mirror image of Fama-French (1993) results and LMH is used instead of HML [See [Sehgal.S et.al \(2012\)](#)].

RMW

RMW stands for robust minus weak and it mimics the risk in relation to company profitability. RMW is estimated as under:

$$RMW = (S/R + B/R)/2 - (S/W + B/W)/2 \quad (9)$$

CMA

CMA stands for conservative minus aggressive and it mimics the risk with regard to company investment. CMA is arrived as noted below:

$$CMA = (S/C + B/C)/2 - (S/A + B/A)/2 \quad (10)$$

LcMHc

LcMHc¹² stands for low compensation minus high compensation and it is the mimicking portfolio which shows the risk involved with the company human capital. it is calculated as shown below:

$$LcMHc = (S/Lc + B/Lc)/2 - (S/Hc + B/Hc)/2 \quad (11)$$

The study uses three regression models:

Fama-French three factor model

$$R_{Pt} - R_{Ft} = a + b (R_{Mt} - R_{Ft}) + s SMB_t + l LMH_t + e_t \quad (12)$$

where,

SMB mimics the risk factor in returns considering size

LMH mimics the risk factor in returns considering value

s and l are the portfolio's responsiveness to (sensitivity coefficients) SMB and LMH factors respectively.

Fama-French Five factor model

$$R_{Pt} - R_{Ft} = a + b (R_{Mt} - R_{Ft}) + s SMB_t + l LMH_t + r RMW_t + c CMA_t + e_t \quad (13)$$

where,

SMB mimics the risk factor in returns considering size

LMH mimics the risk factor in returns considering value

RMW mimics the risk factor in returns considering profitability

CMA mimics the risk factor in returns considering investment

S, l, p and t are the portfolio's responsiveness to (sensitivity coefficients) SMB, LMH, RMW and CMA factors respectively.

Six factor model with human capital

$$R_{Pt} - R_{Ft} = a + b (R_{Mt} - R_{Ft}) + sSMB_t + lLMH_t + rRMW_t + c CMA_t + lLcMHct + e_t \quad (14)$$

where,

SMB mimics the risk factor in returns considering size

LMH mimics the risk factor in returns considering value

¹³LcMHc is calculated in similar way that of other mimicking portfolios used in the study.

RMW mimics the risk factor in returns considering profitability

CMA mimics the risk factor in returns considering investment

LcMHc mimics the risk factor in returns considering human capital

S, l, p, t and h are the portfolio's responsiveness to (sensitivity coefficients) SMB, LMH, RMW, CMA, and LcMHc factors respectively.

Explanatory variables

Table 1 shows the descriptive statistics for the independent variables. The results show that the monthly average market premium is 0.9% ($t=1.461$) while monthly size premium is found to be 1.3% ($t=5.120$). The monthly value premium is of 0.6% ($t=1.650$) while profitability premium is estimated to be 0.2% ($t=0.710$) per month. Then premium for investment is 0.3% ($t=1.179$) per month. Finally, human capital premium 0.2% ($t=0.910$) for the respective mimicking portfolios. The average market premium of 0.9%; size premium of 1.3% and value premium of 0.6% seems large in terms of statistics and practical terms from the investment perspective. The results are echoing the previous studies of [Fama-French \(1993, 2015\)](#). The average profitability premium of 0.2%, investment premium of 0.3% and human capital premium of 0.2% seems much lower than the previous three (market, size and value) and for all the cases standard errors are less.

Insert Table 1

Correlation matrix for the explanatory variables is shown in the Table 2. The results shows negative correlation between the following variables: 'market and size', 'profitability and investment'; 'size and profitability'; 'investment and profitability'. The finding are in line with the findings of Fama-French (2015) in US context. Market is positively related to the value and human capital factor; size is positively related to the market as it is obvious beta for the small companies are higher. Value factor is negatively related to the profitability factor and positively related to invest and human capital factor, theoretically it is obvious. Human capital factor is positively related to the all factors except profitability and it is obvious if you spent more on human capital profits will be less.

Insert Table 2

Empirical results

Figure 1 shows that small stocks earn significantly higher returns than big stocks and this phenomenal performance is pervasive in all the portfolios across the combinations (MC_PB, MC_ROE, MC_TA and MC_HC). Hence this is the manifestation of size, value, investment, and human capital effects.

Insert Figure 1

Table 3, Panel A shows mean excess returns on double sorted portfolios constructed on MC_PB, MC_ROE, MC_TA and MC_HC.

MC_PB:

Mean excess returns on MC_PB sorted portfolio As stated in the previous section, similar observation can be drawn that size effect dominates in Indian market and our result is just the mirror image of the [Fama-French \(1993, 2015\)](#) studies since we have used P/B ratio instead of B/M for value. Portfolio P1 with combination of small size and low value stocks yields a monthly mean excess return of 4.6% per month, which is 3.28 times higher than the monthly mean excess return of portfolio P5 (last portfolio in the first row) with combination of big size and low value stocks and 5.1 times higher than the monthly mean excess return of portfolio P25 with combination of big size and high value stocks. In every column by holding the value constant the mean excess return of the portfolio falls from top to bottom and this is known as the size effect. Similar observation can be made in first row by holding the size constant the mean excess return of portfolio falls from left to right and this is known as the value effect. The findings are similar to the findings of [Fama-French \(1993, 2015\)](#). Further the portfolio mean excess return pattern shows that the value effect weakens successively from row second to last row.

MC_ROE:

Strong size effect observed in the mean excess return pattern for MC_ROE crosses portfolios. Portfolio P1 with combination of small size and weak profitability stocks yields a monthly mean excess return of 4.2% per month, which is 6 times higher than the monthly mean excess return of portfolio P5 (last portfolio in the first row) with combination of big size and weak profitability stocks and 3.2 times higher than the monthly mean excess return of portfolio P25 with combination of big size and robust profitability stocks. In every column by holding the profitability constant the mean excess return of the portfolio falls from top to bottom. No

much clear pattern observed by holding the size constant the mean excess return of portfolio shows no clear pattern on first three rows moving from left to right but last two rows with big size portfolios shows weak profitability effect. The findings are closely similar to the findings of [Novy-Marx \(2013\)](#) & [Fama-French \(2015\)](#) study results.

MC_TA:

Similar to MC_ROE crosses portfolio returns, Strong size effect observed in the mean excess return pattern for MC_TA cross portfolios. Portfolio P1 with combination of small size and conservative investment stocks yields a monthly mean excess return of 4.1% per month, which is 3.42 times higher than the monthly mean excess return of portfolio P5 (last portfolio in the first row) with combination of big size and conservative investment stocks and 4.55 times higher than the monthly mean excess return of portfolio P25 with combination of big size and aggressive investment stocks. In every column by holding the investment constant the mean excess return of the portfolio falls from top to bottom. In rows by holding the size constant the mean excess return of portfolio shows no clear pattern.

MC_HC:

In details in the literature review section of this paper we discussed about the importance of human capital in asset pricing and missing it may lead to the serious problems [Campbell \(1992, 1996\)](#). So, similar to the other factors based on size and human capital 25 portfolios are constructed to determine any human capital effect. Similar to the previous cases size effect predominantly observed in the mean excess return of MC_HC cross portfolios. Portfolio P1 with combination of small size and low compensation stocks yields a monthly mean excess return of 3.1% per month, which is 3.44 times higher than the monthly mean excess return of portfolio P5 (last portfolio in the first row) with combination of big size and low compensation stocks and 4.43 times higher than the monthly mean excess return of portfolio P25 with combination of big size and high compensation stocks. In every column holding the human capital constant the mean excess return of the portfolio falls from top to bottom and this effect is known as the size effect. Similar observation made in every row except the first row by holding the size constant the mean excess return of portfolio falls from left to right and we termed this effect as the human capital effect.

Insert Table 3

Carefully looking into the mean excess return pattern of the portfolios formed from MC_HC cross details several important aspects are explained in later section. It is true that in every row except the first row by holding the size constant the mean excess return of 1st portfolio with low compensation is higher than the extreme right 5th portfolio with high compensation but in between portfolios slopes changes frequently with different frequency and shows non-linear or complex return pattern. Complex pattern of mean excess return of the portfolio points to the fact that compensation (salary and wages) are very important component in estimating the portfolio returns. Visual 3-D surface images for different cross portfolios are shown in the figure 2.

Insert Figure 2

Asset pricing results

Table 4 details about the Fama-French three factor (FFTF) regression results for MC_PB, MC_ROE, MC_TA and MC_HC cross portfolios.

MC_PB

Panel A of Table 4 represents the FFTF regression result for the MC_PB cross portfolios. A model is said to be a good that can captures all alpha values equal to zero. The alpha value ranges from 0.001 to 0.014 with an average alpha value of 0.004 for all 25 portfolios. Four portfolios out of twenty five portfolios are found to be statistically significant (alpha values where $t(a)$ is more than 1.96) including the microcap stocks with low value is not captured by the model. Similar to the [Fama-French \(2015\)](#) study our results also find that microcap is the problem.

MC_ROE

In case of MC_ROE cross portfolios the alpha value ranges from 0.000 to 0.011 with an average alpha value of 0.005 for all 25 portfolios (see Panel B of table 4). Six portfolios out of twenty five portfolios are found to be statistically significant (alpha values where $t(a)$ is more than 1.96) including the microcap stocks with weak profitability is not captured by the model. [Fama-French \(2015\) main problem lies with the](#) microcap stock portfolio are not captured by the model in US context. Our study results are in similar line with [Fama-French \(2015\)](#) results, unable to capture the microcap stocks in Indian context.

MC_TA

Three factor regressions for MC_TA cross portfolios estimated results are similar to the previous two cases. The average alpha value calculated is 0.004 for all portfolios while the alpha values lies between the values 0.000 to 0.011 (See Panel C of table 4). Five portfolios found statistically significant (alpha values where $t(a)$ is more than 1.96) including the microcap stocks with conservative investment is not captured by the model. Similar to the [Aharoni, Grundy, and Zeng \(2013\)](#), [Fama-French \(2015\)](#) study our results also find that microcap is the problem.

MC_HC

Three factor regressions estimates for MC_HC cross portfolios shows three factor model unable to capture all alpha values equal to zero. Three portfolios out of twenty five portfolios found statistically significant (alpha values where $t(a)$ is more than 1.96) is not captured by the model. The microcap stocks combination of small size and low compensation is well captured by the model.

R^2 Value for FTF models

Panel E represents the R^2 values of the FTF regressions. The average value of R^2 for MC_PB, MC_ROE, MC_TA & MC_HC are 82.7%, 81.1%, 81.7% and 79% respectively.

Insert Table 4

Table 5 details about the Fama-French Five factor (FFFF) regression results for MC_PB, MC_ROE, MC_TA and MC_HC cross portfolios.

MC_PB

Panel A of table 5 represents the FFFF regression results for the MC_PB cross portfolios. The alpha value ranges from 0.000 to 0.015 with an average alpha value of 0.004 for all 25 portfolios. Four portfolios found statistically significant (alpha values where $t(a)$ is more than 1.96) including the microcap stocks with low value is not captured by the model. Results of FFFF are similar to the FTF results and no significant improvement observed. The market slopes are highly positive and SMB slopes are positive for small stocks whereas slightly negative for the few big stock portfolios. FFFF model gives more information about the microcap stocks with low value, the portfolio's slope sign of LMH (0.699, $t=4.061$) and CMA (-0.487, $t=-2.697$) are similar to the other low value portfolios but RMW (0.014, $t=0.073$) slope is different and changes to negative with increase in size. First portfolio

combination of microcap stocks and low value stocks are most dominated by the microcap whose returns behave like those low value firms with aggressive investment that have robust profitability. Since much of the results has similar story to the mean returns for the different cross of MC with other factor but it is important to see whether the factor variable slopes line up with the LMH, RMW and CMA factors. Summary statistics in table 3 will answer often but not always. Similarly regression slopes used to align with the characteristics used to form left hand side portfolios but not with the other factors. For example low value portfolios shows strong positive slope pattern for LMH whereas high value portfolios shows strong negative slope pattern for LMH. MC_PB cross is not based on the investment factor but still it is observed that low P/B value portfolios shows weak negative slope pattern for CMA whereas high P/B value portfolios shows comparatively high negative slope pattern for CMA. It lines up with the evidence that low P/B value stocks have conservative investment and high P/B value stocks have aggressive investment pattern. Similarly is also observed in case of profitability see table 3. It will be high expectation to map univariate characteristics with multivariate regression slope but what is important is whether LMH, RMW & CMA slopes line up better with future value than past. Just to explain previously we have explained that “The microcap stocks with low value portfolio are most dominated by the microcap whose returns behave like those low value firms with conservative investment with marginal profits” but they are more profitable than other microcap portfolios see table 3.

Panel B of table 5 represents the FFFF regression result for the MC_ROE cross portfolios. The alpha value ranges from 0.000 to 0.011 with an average alpha value of 0.004 for all 25 portfolios. Four portfolios out of twenty five portfolios found statistically significant (alpha values where $t(a)$ is more than 1.96) including the microcap stocks with low profitability is not captured by the model. Results slightly improved for FFFF than the FFTF results and the FFFF intercepts shows close pattern to the mean excess return (see table 3). FFFF model gives more information about the microcap stocks with low value, the portfolio's slope of LMH (0.278, $t=1.677$) and RMW (-0.405, $t=-2.127$) are similar to the other low value portfolios but CMA (0.036, $t= 0.207$) slope is different. The microcap stocks with low profitability portfolio are most dominated by the microcap whose returns behave like those unprofitable firms with conservative investment. Though FFFF able to capture more alpha value close to zero for more portfolios than FFTF but still has big problem with microcaps, our results contradict to [Fama-French \(2015\)](#) finding.

Panel C of table 5 represents the FFFF regression result for the MC_TA cross portfolios. The alpha value ranges from 0.000 to 0.010 with an average alpha value of 0.004 for all 25 portfolios. Five portfolios out of twenty five portfolios found statistically significant (alpha values where $t(a)$ is more than 1.96) including the microcap stocks with conservative investment is not captured by the model. Results of FFFF are similar to the FFTF results and no improvement observed and big problem lies with microcaps, our results is similar to [Fama-French \(2015\)](#) finding. Biggest problem lies with the unprofitable microcap with conservative investment. .

Panel D of table 5 represents the FFFF regression result for the MC_HC cross portfolios. The alpha value ranges from 0.000 to 0.016 with an average alpha value of 0.004 for all 25 portfolios. Three portfolios out of twenty five portfolios found statistically significant (alpha values where $t(a)$ is more than 1.96). Results of FFFF are similar to the FFTF results and no improvement observed but able to capture the microcap stocks with low compensation. The biggest problem of [Fama-French \(2015\)](#) is unable to capture the microcap with high growth due to high negative CWA and RMW slopes, which is well captured by the MC_HC portfolios because it do not isolates firms that behaves like firms that invest a lot in spite of low profitability.

R² Value for FFFF models

Panel E represents the R² values of the FFTF regressions. The average value of R² for MC_PB, MC_ROE, MC_TA & MC_HC are 84%, 82.8%, 83.3% and 80.1% respectively. Though the R² value increases in case of FFFF model but study results find that FFTF is more robust than FFFF in Indian market and is contradictory to the [Fama-French \(2015\)](#) and favours the [Hou, Xue, Zhang \(2014\)](#); [Clarke \(2016\)](#); [Chiah, Chai, Zhong& Li \(2016\)](#) results of failure of FFFF. The study finds that estimates of Fama-French factor models yield different model for regional and country level and result supports [Griffin \(2002\)](#), [Fama and French \(2012, 2015c\)](#) findings.

Insert Table 5

Redundant variable

FFFF model shows no much significant improvement from FFTF in capturing the risk return relationship of portfolios. Next study try to identify if any redundant variable is there that can be dropped similar in line of [Fama-French \(2015\)](#) we run four factor regressions to explain

the average returns on the fifth and the regression results are shown in table 6. Fama-French (2015) found HML factor redundant in our result we found LMH factor is not redundant but CMA (investment factor) is redundant no study in Indian context has reported the same previously. In Indian context HML or value factor found to be empirically important whereas CMA or investment factor is redundant see table 6. In the CMA regressions, the large average return is absorbed by RMW and SMB. The RMW slopes are highly negative which in line with that the low profitability firm tend to have high investment but the SMB slopes are strongly positive which says that controlling for other factors, aggressive stocks behave like stocks with small size, even though unconditionally big stock tend to be more profitable. Dropping the CMA (investment factor) we find the similar results as the reported results with CMA factor in the model (See table 5) for all cross. We can easily drop the CMA investment factor but there is investment premium in average returns that is often targeted by money managers. So similar to [Fama-French \(2015\)](#) we define the CMAO (orthogonal CMA) as the sum of intercepts and residuals from the regression of CMA on $R_m - R_f$, SMB, LMH& RMW. Substituting CMAO in the equation 13 we get the below equation 15.

$$R_{Pt} - R_{Ft} = a + b (R_{Mt} - R_{Ft}) + s SMB_t + l LMH_t + p RMW_t + t CMAO_t + e_t \quad (15)$$

Both the intercepts and residuals of five factor regression are with equation 13 and equations 15 are same. As expected the regressions result with new modified five factor model with CMAO do not reflect any improvement from reported five factor model with CMA results in table 5 as the slopes of CMA are same. Results for five factor regressions with CMAO not reported in the study but can be available on request. We have also tested five factor regressions by dropping the CMAO factor with $R_m - R_f$, SMB, LMH, RMW and LcMHc but no significant improvement in result observed from the previous three, four and five factor regressions results and not able to capture the portfolio one returns of MC_PB, MC_ROE and MC_TA cross portfolios.

Insert Table 6

For model comparison then we add an extra variable of HC (Human Capital) to the equation 15 and it becomes equation 16. Then we run the regression on the set of left hand side portfolios and results are shown in the table 7.

$$R_{Pt} - R_{Ft} = a + b (R_{Mt} - R_{Ft}) + s SMB_t + l LMH_t + p RMW_t + t CMAO_t + h LcMHct + e_t \quad (16)$$

Table 7 details about the six factor regression results for MC_PB, MC_ROE, MC_TA and MC_HC cross portfolios.

MC_PB

Six factor regression estimates for MC_PB crosses portfolios found three portfolios out of twenty five portfolios found statistically significant (alpha values where $t(a)$ is more than 1.96) including the microcap stocks with low value is not captured by the model (See Panel A of table 7). Results of six factor regressions are still better than both the FFFF and FFTF results and average alpha value also decreases. Though six factor model do not able to capture the alpha value equal to zero for the extreme portfolio with low value microcaps but reveals more information than five factor model. Six factor model gives more information about the microcap stocks with low value than FFFF, the portfolio's slope of LMH (0.732, $t=4.357$) is similar but LcMHc (0.521, $t=-3.296$) and RMW (0.316, $t= 1.502$) positive slope changes to negative with increase in size. FFFF models finds that First portfolio combination of microcap stocks and low value stocks are most dominated by the microcap whose returns behave like those low value firms with aggressive investment that have robust profitability but six factor add addition information of low human capital investment. In a nut shell first portfolio are mostly dominated by the microcap whose returns behave like those low value firms with aggressive investment in assets & low human capital investment that have robust profitability. Hence, low value microcaps that invest lot in company asset growth but invest very low in human asset growth earns high profits are not captured by the asset pricing models. In Indian market microcap is dominated by low value stocks companies that are highly profitable invest aggressively for asset growth but invest much lesser for human growth. So, these are the companies with low values that overuse the human capital and pays low compensation for making profits.

MC_ROE

Six factor regression for the MC_ROE cross portfolios unable to capture three portfolios that are statistically significant (alpha values where $t(a)$ is more than 1.96) including the microcap stocks portfolio. Though the six factor regressions estimated results are much better than both the FFFF and FFTF estimates. Average alpha value decreases for six factor regression estimates but unable to capture the alpha value equal to zero for extreme left portfolio with small size and conservative investment. First portfolio is mostly dominated by the microcap whose returns behave like those unprofitable firms with conservative investment in assets &

human capital investment. So, these are the unprofitable companies stocks that invest very low for its asset and human capital growth, that are not captured by the model.

MC_TA

Estimates of six factor regressions are far better than both the FFFF and FFTF results in case of MC_TA. Interestingly six factor regression is able to captures almost all alpha value equal to zero including the microcap with small size and low investment well captured by the model. Five factor model biggest problems are to capture the returns of the unprofitable microcap with conservative investment which is well captured by the six factor model. The high negative coefficient of the RMW and CMAO factor reduced significantly and returns are more explained by the positive coefficient of human capital.

MC_HC

For MC_HC crosses six factor regressions are much better than both the FFFF and FFTF estimates and only one portfolio is not captured. Interestingly microcap with small size and low human capital growth which was problem for both FFTF and FFFF are captured by six factor model. The high negative coefficient of the RMW and CMAO factor reduced significantly and returns are more explained by the positive coefficient of human capital.

R² Value for Six factor model with HC

Panel E represents the R² values of the FFTF regressions. The average value of R² for MC_PB, MC_ROE, MC_TA & MC_HC are 85%, 84%, 84% and 81% respectively.

Insert Table 7

In a nutshell the overall results are documented in the table 8.

Empirical result confirms the total failure of Fama-French Five factor model similar to [Hou, Xue, Zhang \(2014\)](#); [Clarke \(2016\)](#); [Chiah, Chai, Zhong& Li \(2016\)](#) findings. Further study supports [Griffin \(2002\)](#), [Fama and French \(2012, 2015c, 2016\)](#) findings of Fama-French factor models yield different model for regional and country level. Introduction of the sixth factor as human capital significantly improved the regression results than three and five factor regressions. Introducing human capital factor reduces the biggest problem of the high negative coefficient of the RMW and CMAO factors. Unfortunately the puzzle of “microcap with low value stocks that are highly profitable invest aggressively for asset growth but invest

much lesser for human growth” and “microcap with unprofitable stocks whose returns behave like those low value firms with conservative investment” are not still not answered by the six factor model.

Insert table 8

Residual graphs

Residual graphs for three, five and six factor regression for Portfolio 1 & Portfolio 25 are shown in the figure 3. Less the peaks and close to zero better is the model, peaks are much closer to zero in case of the six factor model than FFTF & FFFF both for portfolio 1 and portfolio 25.

Insert Figure 3

Model Performance Test (GRS-Test)

GRS¹³ test is model performance test designed by [Gibbons, Ross &Shanken \(1989\)](#). The GRS test results for all the factor models are shown in the table 9. All three factor models are rejected by the GRS test and it is consistent with the recent global findings of [Fama-French \(2012, 2015\)](#) and do not support [Connor &Sehgal \(2003\)](#) findings in Indian context. Except for MC_PB & MC_ROE cross portfolios all other five factor models are accepted by GRS test and except MC_PB cross portfolios all other six factor models are accepted by GRS test and contradict to [Fama-French \(2012, 2015\)](#) finds in US context. [Fama-French \(2015\)](#) study argue that lower GRS F-Statistics signifies six factor model a better model than five factor model and it is consistent with our results (see table 9) and that confirms the superiority of six factor model in all cases over all both the three and five factor models.

Insert table 9

Conclusion

There are average returns patterns related to the Size, value, profitability, investment and human capital in Indian capital market. GRS test easily rejects both the three factor model and five factor model directed at capturing these average return patterns for size, value, profitability and investment. The study estimated that the both three and five factor model has

¹⁴GRS is computed in MS excel as explained by the Karl Diether (2001) university of Chicago in “GRS review”. It can be accessed from the following link:
<http://faculty.chicagobooth.edu/eugene.fama/teaching/Reading%20List%20and%20Notes/GRS.pdf>

the similar explanatory power and R squared value are nearly similar. In Indian context study results find LMH factor as an important factor unlike the [Fama-French \(2015\)](#) findings in US context but CMA factor as the redundant factor. In case of parsimony issue our study results suggest that investment (CMA) is a redundant factor and average returns due to CMA is fully captured on exposure to the Rm-Rf, SMB, LMH and RMW factors. The investors whose sole interest is to capture the abnormal returns (regression intercepts/ alpha value), study results suggest that four factor model without CMA factor yields similar results as of the five factor model. Five factor model will be of interest for the investors that have choice to tilt towards size, value, profitability and investment premiums, study suggests using the CMAO factor as a replacement for the CMA factor in the five factor model. Moreover study result suggests the complete failure of the five factor model over three factor model in Indian context similar to [Hou, Xue, Zhang \(2014\)](#); [Clarke \(2016\)](#); [Chiah, Chai, Zhong & Li \(2016\)](#) studies in international context. Further study also supports the [Griffin \(2002\)](#), [Fama and French \(2012, 2015c, 2016\)](#) findings of Fama-French factor model are not global models.

Most of the previous studies have ignored the human capital factor in asset pricing models. For the first time our study introduced human capital as the sixth factor in Indian context and find interesting results. Our six factor model with human capital able to capture most of the average returns patterns related to the Size, value, profitability, investment and human capital in Indian capital market than both the three and five factor models. Almost all portfolios return captured by the six factor model for MC_ROE (Size and profitability), MC_TA (Size and Investment) & MC_HC (Size and Human Capital) portfolios and the model is not rejected by the GRS test. Similarly for MC_PB (Size and value) portfolios six factor model yields better results than both the three and five factor model. Human capital as the sixth factor significantly improved the regression results and model efficiency than both three and five factor models. Human capital factor variable in the regressions reduces the biggest problem of the high negative coefficient of the RMW and CMAO factors. Unfortunately the puzzle of “microcap with low value stocks that are highly profitable invest aggressively for asset growth but invest much lesser for human growth” and “microcap with unprofitable stocks whose returns behave like those low value firms with conservative investment” are not answered by the six factor model. The study finds that the human capital factor as one of the important factor affecting the portfolio returns in Indian context and been ignored by previous all studies. Investment decision based on the human capital factor of the company adds much more reliable information about the value of the company and helps the investor

to take rational decision is one of the main message here. Though our six factor model avail to capture the most of the abnormal return of the microcaps but still asset pricing model have greatest problems with microcaps as similar to global findings of [Fama-French \(1993, 2012, 2015, 2016\)](#).

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Table 1: Descriptive statistics for the independent variables

	RM	SMB	LMH	RMW	CMA	LcMHc
Mean returns	0.009	0.013	0.006	0.002	0.003	0.002
Standard deviations	0.071	0.031	0.045	0.043	0.026	0.033
T-Statistics	1.461	5.120	1.650	0.710	1.179	0.910

Table 2: Correlation matrix for explanatory variables

	RM	SMB	LMH	RMW	CMA	LcMHc
RM	1					
SMB	0.090	1				
LMH	0.351	0.337	1			
RMW	-0.434	-0.263	-0.723	1		
CMA	-0.124	0.354	0.287	-0.369	1	
LcMHc	0.424	0.144	0.509	-0.648	0.178	1

Figure 1: Radar graph for 25 portfolios mean returns formed on MC_PB, MC_ROE, MC_TA and MC_HC

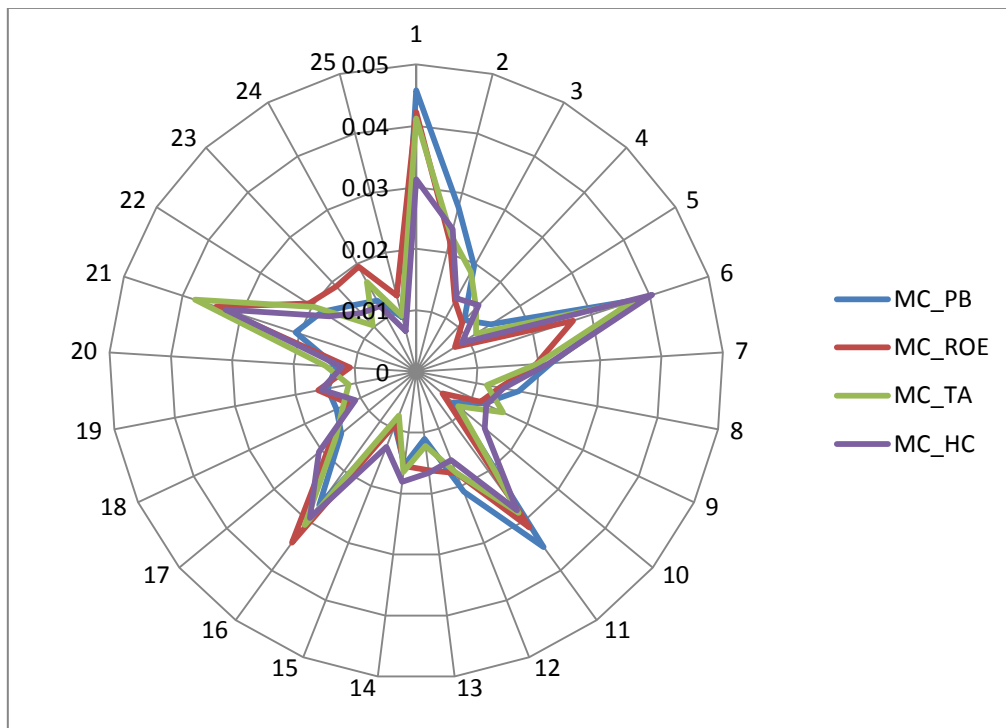


Table No. 3: Summary statistics of 25 portfolios formed on MC_PB, MC_ROE, MC_TA and MC_HC

Panel A (Mean excess returns)

MC_PB						MC_ROE					
	Low	2	3	4	High		Low	2	3	4	High
Small	0.046	0.038	0.035	0.027	0.021	Small	0.042	0.027	0.031	0.034	0.034
2	0.028	0.022	0.021	0.016	0.018	2	0.022	0.020	0.018	0.018	0.021
3	0.020	0.017	0.011	0.014	0.015	3	0.013	0.014	0.016	0.012	0.019
4	0.011	0.012	0.015	0.015	0.013	4	0.011	0.012	0.016	0.016	0.019
Big	0.014	0.008	0.009	0.013	0.009	Big	0.007	0.006	0.009	0.011	0.013

MC_TA						MC_HC					
	Low	2	3	4	High		Low	2	3	4	High
Small	0.041	0.037	0.028	0.031	0.038	Small	0.031	0.040	0.028	0.029	0.033
2	0.023	0.020	0.018	0.017	0.020	2	0.024	0.022	0.015	0.021	0.017
3	0.018	0.012	0.012	0.013	0.010	3	0.014	0.014	0.017	0.011	0.013
4	0.014	0.016	0.016	0.011	0.017	4	0.015	0.013	0.018	0.016	0.012
Big	0.012	0.009	0.008	0.015	0.009	Big	0.009	0.015	0.013	0.012	0.007

Panel B (Standard deviations)

MC_PB						MC_ROE					
	Low	2	3	4	High		Low	2	3	4	High
Small	0.100	0.102	0.093	0.084	0.085	Small	0.107	0.096	0.095	0.090	0.091
2	0.100	0.097	0.080	0.082	0.078	2	0.107	0.096	0.082	0.083	0.080
3	0.104	0.102	0.080	0.087	0.075	3	0.108	0.100	0.092	0.081	0.079
4	0.114	0.097	0.107	0.087	0.067	4	0.107	0.100	0.089	0.083	0.068
Big	0.107	0.103	0.086	0.086	0.066	Big	0.103	0.100	0.083	0.079	0.077

MC_TA						MC_HC					
	Low	2	3	4	High		Low	2	3	4	High
Small	0.104	0.106	0.094	0.092	0.099	Small	0.116	0.098	0.091	0.091	0.084
2	0.098	0.093	0.087	0.084	0.094	2	0.105	0.095	0.088	0.081	0.083
3	0.093	0.084	0.092	0.092	0.095	3	0.103	0.088	0.090	0.093	0.082
4	0.093	0.086	0.087	0.085	0.110	4	0.104	0.074	0.086	0.082	0.088
Big	0.078	0.072	0.080	0.084	0.095	Big	0.094	0.103	0.082	0.075	0.079

Panel C (T-Statistics)

MC_PB						MC_ROE					
	Low	2	3	4	High		Low	2	3	4	High
Small	5.566	4.562	4.635	3.907	2.983	Small	4.821	3.401	4.051	4.667	4.613
2	3.366	2.764	3.215	2.357	2.858	2	2.494	2.484	2.658	2.664	3.158
3	2.300	2.031	1.690	2.016	2.430	3	1.481	1.692	2.145	1.826	2.939
4	1.225	1.539	1.741	2.061	2.412	4	1.251	1.405	2.126	2.374	3.495
Big	1.642	0.931	1.340	1.857	1.693	Big	0.886	0.674	1.331	1.670	2.028

MC_TA						MC_HC					
	Low	2	3	4	High		Low	2	3	4	High
Small	4.847	4.283	3.684	4.112	4.675	Small	3.285	5.016	3.750	3.948	4.732
2	2.850	2.579	2.494	2.469	2.560	2	2.779	2.796	2.157	3.096	2.457
3	2.421	1.693	1.627	1.732	1.330	3	1.616	2.007	2.246	1.439	1.948
4	1.869	2.200	2.313	1.614	1.842	4	1.737	2.083	2.569	2.350	1.689
Big	1.817	1.500	1.174	2.135	1.214	Big	1.144	1.712	1.970	1.975	1.053

Figure 2: Visual 3-D surface images for 25 portfolios mean returns formed on MC_PB, MC_ROE, MC_TA and MC_HC

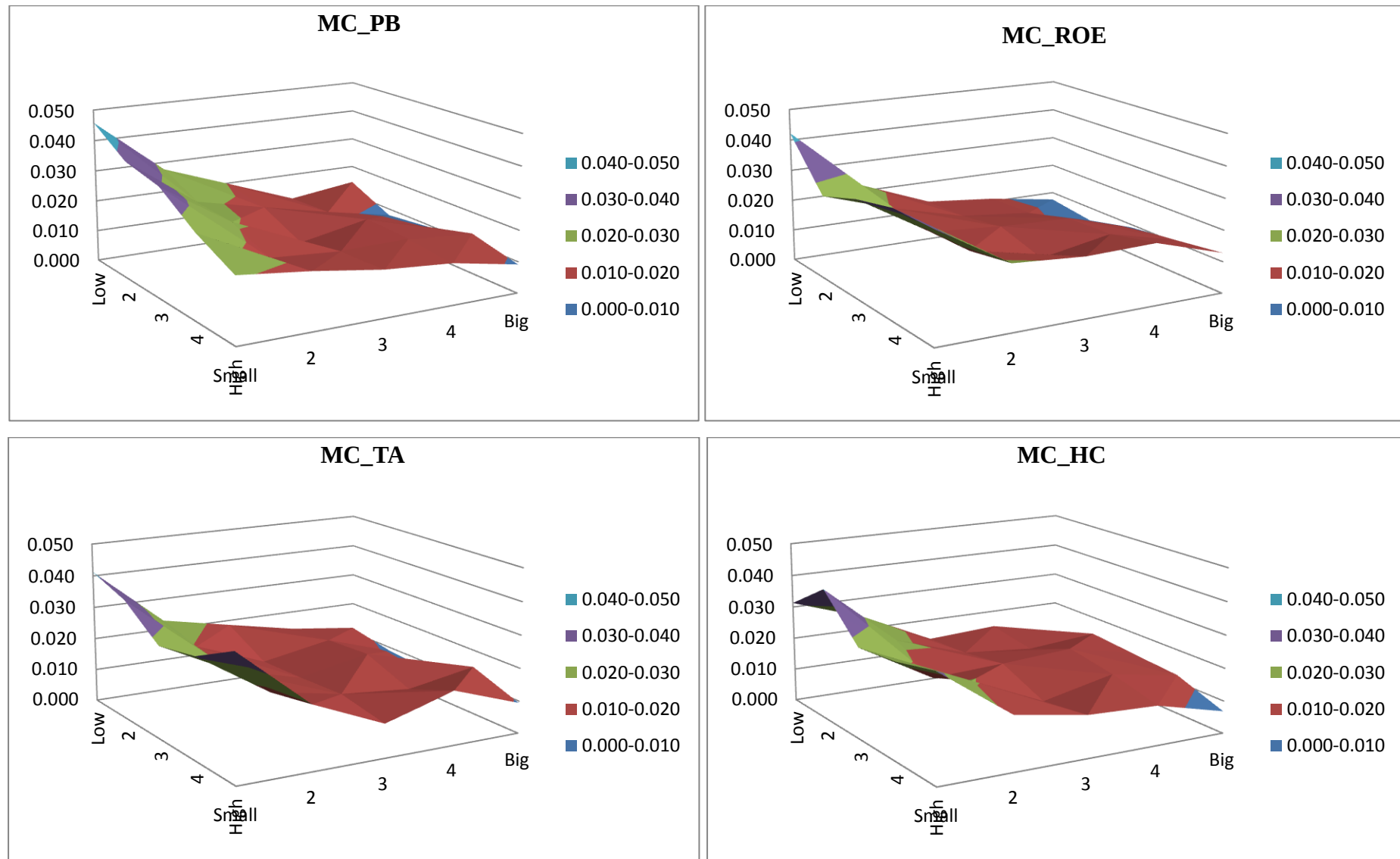


Table 4 Regression results of Fama-French Three Factor Model for 25 portfolios

$$R_{Pt} - R_{Ft} = a + b (R_{Mt} - R_{Ft}) + s \text{ SMB}_t + l \text{ LMH}_t + e_t$$

Panel A (MC_PB)

a						b				
	Low	2	3	4	High	Low	2	3	4	High
Small	0.014	0.007	0.011	0.001	-0.003	0.957	1.159	1.149	1.050	1.108
2	-0.002	0.001	0.000	-0.006	0.002	0.987	1.078	0.943	1.017	1.032
3	-0.004	-0.005	-0.005	-0.002	0.003	1.085	1.217	0.931	1.166	1.003
4	-0.002	0.001	0.004	0.002	0.005	1.054	0.977	1.302	1.139	0.901
Big	0.004	-0.002	0.002	0.006	0.001	0.986	1.126	1.042	1.120	0.924

s						l				
	Low	2	3	4	High	Low	2	3	4	High
Small	1.469	1.400	1.011	1.252	1.200	0.778	0.554	0.272	0.107	-0.132
2	1.207	0.735	0.834	0.906	0.661	0.894	0.463	0.331	0.184	-0.134
3	0.675	0.646	0.482	0.491	0.337	0.912	0.503	0.318	-0.055	-0.134
4	-0.218	-0.116	-0.080	0.282	0.216	1.172	0.689	0.280	-0.049	-0.290
Big	-0.372	-0.300	-0.232	-0.163	0.122	1.076	0.689	0.296	-0.065	-0.288

t(a)						t(b)				
	Low	2	3	4	High	Low	2	3	4	High
Small	3.184	1.999	2.952	0.331	-0.955	15.358	21.931	22.110	22.111	21.793
2	-0.461	0.121	0.080	-1.665	0.495	18.614	16.485	20.142	21.195	21.706
3	-1.108	-1.274	-1.450	-0.494	0.930	21.843	23.029	18.878	25.223	23.337
4	-0.498	0.296	0.898	0.472	1.569	20.431	17.129	23.066	22.676	22.076
Big	1.332	-0.733	0.747	2.133	0.656	21.873	28.147	31.985	27.909	29.173

t(s)						t(l)				
	Low	2	3	4	High	Low	2	3	4	High
Small	10.882	12.230	8.986	12.179	10.898	8.228	6.901	3.449	1.486	-1.709
2	10.504	5.194	8.221	8.717	6.419	11.102	4.668	4.656	2.520	-1.859
3	6.275	5.644	4.515	4.908	3.622	12.097	6.268	4.242	-0.786	-2.050
4	-1.954	-0.941	-0.653	2.598	2.448	14.971	7.964	3.265	-0.638	-4.688
Big	-3.814	-3.465	-3.282	-1.880	1.772	15.730	11.354	5.977	-1.071	-5.983

Panel B MC_ROE

a						b				
	Low	2	3	4	High	Low	2	3	4	High
Small	0.009	0.002	0.005	0.008	0.007	1.035	1.044	0.989	1.045	1.106
2	-0.005	-0.002	0.000	-0.003	0.003	1.060	1.081	1.022	1.071	1.036
3	-0.011	-0.001	0.000	-0.003	0.006	1.122	1.141	1.126	0.999	1.034
4	-0.005	-0.002	0.006	0.008	0.010	1.153	1.031	1.002	0.957	0.923
Big	-0.003	-0.005	0.000	0.001	0.006	1.171	1.180	1.018	1.050	1.014

s						l				
	Low	2	3	4	High	Low	2	3	4	High
Small	1.645	1.047	1.191	1.337	1.513	0.592	0.436	0.430	0.113	-0.220
2	1.032	0.756	0.648	1.014	0.770	0.842	0.489	0.107	-0.206	-0.095
3	0.773	0.199	0.410	0.494	0.337	0.720	0.453	0.256	0.150	-0.044
4	0.165	0.040	-0.095	-0.118	0.245	0.651	0.746	0.402	0.181	-0.286
Big	-0.208	-0.228	-0.111	0.095	-0.035	0.581	0.495	0.250	-0.058	-0.203

t(a)						t(b)				
	Low	2	3	4	High	Low	2	3	4	High
Small	2.064	0.516	1.289	2.112	1.775	16.439	17.758	16.698	19.408	19.312
2	-1.314	-0.567	0.087	-0.728	0.821	18.705	20.370	22.037	21.071	21.947
3	-2.370	-0.281	-0.032	-1.029	2.036	17.656	19.921	22.400	20.793	23.633
4	-1.130	-0.565	1.598	2.208	3.713	18.638	18.201	19.495	17.486	23.355
Big	-1.000	-1.595	0.095	0.382	1.986	24.347	28.398	27.026	27.106	23.753

t(s)						t(l)				
	Low	2	3	4	High	Low	2	3	4	High
Small	11.826	8.059	9.095	11.231	11.953	5.951	4.697	4.598	1.328	-2.434
2	8.246	6.448	6.322	9.029	7.380	9.413	5.829	1.459	-2.559	-1.275
3	5.510	1.572	3.695	4.657	3.484	7.172	5.007	3.228	1.975	-0.644
4	1.208	0.316	-0.840	-0.978	2.802	6.659	8.336	4.945	2.094	-4.576
Big	-1.962	-2.479	-1.330	1.112	-0.375	7.652	7.543	4.197	-0.955	-3.010

Panel C MC_TA

a						b				
	Low	2	3	4	High	Low	2	3	4	High
Small	0.010	0.008	0.004	0.007	0.011	1.008	1.070	1.003	0.981	1.031
2	0.000	-0.002	-0.003	0.000	-0.001	1.069	1.061	0.966	0.942	1.040
3	-0.002	-0.003	-0.004	-0.002	-0.007	1.055	0.967	1.148	1.132	1.037
4	0.002	0.004	0.004	0.000	0.004	1.106	1.008	1.024	1.014	1.370
Big	0.000	0.001	-0.001	0.009	-0.001	0.962	0.836	1.050	1.068	1.239

s						l				
	Low	2	3	4	High	Low	2	3	4	High
Small	1.730	1.447	1.159	1.277	1.502	0.250	0.377	0.277	0.047	0.036
2	0.929	0.920	0.898	0.634	0.914	0.418	0.212	0.292	0.300	0.101
3	0.727	0.422	0.492	0.362	0.601	0.386	0.274	0.132	0.151	0.194
4	0.128	0.106	0.145	0.066	0.087	0.165	0.339	0.345	0.310	0.069
Big	0.200	-0.049	-0.060	-0.344	0.017	0.120	0.285	0.078	0.190	-0.014

t(a)						t(b)				
	Low	2	3	4	High	Low	2	3	4	High
Small	2.584	2.097	1.185	2.000	2.462	17.666	18.854	19.456	16.894	16.456
2	0.036	-0.436	-0.999	-0.116	-0.178	21.803	20.485	22.320	20.027	16.785
3	-0.487	-0.942	-1.387	-0.508	-1.445	23.556	19.805	26.311	22.705	15.447
4	0.573	1.129	1.269	-0.043	0.791	19.494	21.689	23.349	24.607	21.507
Big	0.089	0.215	-0.478	3.627	-0.383	23.216	21.202	34.348	31.496	27.140

t(s)						t(l)				
	Low	2	3	4	High	Low	2	3	4	High
Small	13.232	11.125	9.809	9.595	10.459	2.597	3.933	3.177	0.481	0.339
2	8.262	7.746	9.060	5.883	6.436	5.036	2.425	3.987	3.769	0.965
3	7.078	3.773	4.919	3.165	3.904	5.101	3.326	1.788	1.786	1.707
4	0.983	0.993	1.445	0.700	0.595	1.718	4.313	4.653	4.447	0.638
Big	2.103	-0.545	-0.852	-4.425	0.166	1.710	4.274	1.507	3.311	-0.184

Panel D MC_HC

a						b				
	Low	2	3	4	High	Low	2	3	4	High
Small	0.000	0.017	0.003	0.004	0.009	1.242	0.887	0.945	0.984	0.917
2	-0.001	-0.001	-0.004	0.001	0.000	1.191	1.068	0.977	0.874	0.911
3	-0.005	-0.001	0.003	-0.006	-0.001	1.136	1.017	1.083	1.175	0.989
4	0.002	0.002	0.007	0.004	0.002	1.151	0.805	0.928	1.009	0.937
Big	0.002	0.003	0.003	0.005	-0.002	0.991	1.317	1.064	0.973	1.057

s						l				
	Low	2	3	4	High	Low	2	3	4	High
Small	1.515	1.079	1.293	1.495	1.473	0.451	0.560	0.235	-0.068	-0.203
2	1.095	0.946	0.761	0.896	0.598	0.269	0.354	0.333	0.226	0.283
3	0.550	0.454	0.269	0.608	0.321	0.428	0.271	0.236	-0.054	0.234
4	-0.060	0.097	0.108	0.261	-0.082	0.554	0.358	0.375	0.065	0.462
Big	-0.369	-0.002	0.095	-0.092	-0.014	0.413	0.037	-0.069	0.024	-0.019

t(a)						t(b)				
	Low	2	3	4	High	Low	2	3	4	High
Small	0.105	3.596	0.893	0.985	2.208	20.837	13.193	17.094	16.201	15.737
2	-0.125	-0.155	-1.058	0.384	0.099	19.813	22.338	19.857	17.143	16.950
3	-1.154	-0.303	0.809	-1.557	-0.174	18.271	19.667	21.540	22.130	23.512
4	0.572	0.767	1.990	1.106	0.598	19.223	17.649	16.267	21.129	16.666
Big	0.477	0.808	1.134	1.947	-0.804	15.623	24.101	24.547	27.871	31.724

	t(s)					t(l)				
	Low	2	3	4	High	Low	2	3	4	High
Small	10.643	6.719	9.788	10.299	10.585	4.524	4.982	2.543	-0.670	-2.084
2	7.626	8.280	6.479	7.355	4.661	2.671	4.430	4.052	2.648	3.151
3	3.704	3.672	2.243	4.793	3.199	4.117	3.134	2.808	-0.614	3.329
4	-0.419	0.891	0.793	2.289	-0.611	5.535	4.697	3.926	0.816	4.914
Big	-2.437	-0.017	0.916	-1.101	-0.180	3.888	0.409	-0.947	0.402	-0.340

Panel E R²

	MC_PB					MC_ROE				
	Low	2	3	4	High	Low	2	3	4	High
Small	0.775	0.842	0.817	0.812	0.789	0.796	0.781	0.768	0.789	0.765
2	0.836	0.735	0.799	0.799	0.780	0.836	0.821	0.810	0.777	0.794
3	0.866	0.843	0.775	0.833	0.807	0.797	0.806	0.825	0.795	0.819
4	0.881	0.796	0.838	0.806	0.782	0.803	0.813	0.804	0.746	0.801
Big	0.896	0.911	0.915	0.872	0.865	0.872	0.899	0.878	0.858	0.819

	MC_TA					MC_HC				
	Low	2	3	4	High	Low	2	3	4	High
Small	0.814	0.823	0.815	0.753	0.752	0.841	0.716	0.775	0.730	0.710
2	0.846	0.811	0.847	0.808	0.730	0.802	0.845	0.809	0.760	0.748
3	0.858	0.794	0.860	0.820	0.694	0.778	0.790	0.812	0.801	0.840
4	0.770	0.822	0.843	0.854	0.795	0.801	0.769	0.733	0.792	0.753
Big	0.825	0.815	0.909	0.901	0.857	0.722	0.831	0.829	0.868	0.893

Table 5 Regression results of Fama-French Five Factor Model for 25 portfolios

$$R_{Pt} - R_{Ft} = a + b (R_{Mt} - R_{Ft}) + s \text{ SMB}_t + l \text{ LMH}_t + p \text{ RMW}_t + t \text{ CMA}_t + e_t$$

Panel A MC_PB

a						b				
	Low	2	3	4	High	Low	2	3	4	High
Small	0.015	0.009	0.013	0.003	0.000	0.899	1.097	1.108	1.000	1.062
2	0.000	0.005	0.003	-0.003	0.006	0.966	1.003	0.895	0.951	0.958
3	0.001	-0.001	-0.004	0.001	0.003	1.022	1.148	0.918	1.102	0.979
4	-0.001	0.002	0.009	0.004	0.005	1.017	0.938	1.202	1.067	0.869
Big	0.005	0.000	0.002	0.005	0.003	0.959	1.088	1.048	1.099	0.897

s						l				
	Low	2	3	4	High	Low	2	3	4	High
Small	1.592	1.540	1.177	1.418	1.286	0.699	0.223	-0.054	-0.151	-0.558
2	1.188	0.965	0.938	1.097	0.781	0.705	-0.130	0.045	-0.194	-0.546
3	0.755	0.792	0.517	0.636	0.464	0.387	0.024	0.081	-0.318	-0.196
4	-0.037	0.031	0.063	0.474	0.301	1.060	0.496	-0.182	-0.345	-0.369
Big	-0.316	-0.227	-0.283	-0.126	0.186	0.977	0.458	0.313	-0.200	-0.444

p						t				
	Low	2	3	4	High	Low	2	3	4	High
Small	0.014	-0.269	-0.280	-0.191	-0.356	-0.487	-0.294	-0.169	-0.300	-0.008
2	-0.043	-0.687	-0.281	-0.412	-0.502	0.081	-0.264	-0.206	-0.357	-0.315
3	-0.585	-0.564	-0.198	-0.354	-0.076	-0.060	-0.239	0.104	-0.418	-0.271
4	-0.256	-0.325	-0.740	-0.446	-0.128	-0.414	-0.309	-0.532	-0.509	-0.275
Big	-0.235	-0.384	-0.003	-0.220	-0.212	-0.208	-0.176	0.079	-0.092	-0.140

t(a)						t(b)				
	Low	2	3	4	High	Low	2	3	4	High
Small	3.309	2.487	3.477	0.833	-0.040	13.505	19.885	21.089	20.807	20.274
2	0.054	1.023	0.789	-0.884	1.670	16.450	15.450	18.265	20.079	19.779
3	0.206	-0.223	-0.977	0.258	0.911	20.489	21.544	17.646	23.361	21.835
4	-0.396	0.572	2.204	1.198	1.781	18.638	15.472	21.315	21.109	20.204
Big	1.599	0.002	0.801	1.454	1.200	19.865	25.785	30.160	25.580	26.894

t(s)						t(l)				
	Low	2	3	4	High	Low	2	3	4	High
Small	10.972	12.809	10.283	13.540	11.266	4.061	1.562	-0.399	-1.219	-4.115
2	9.285	6.819	8.785	10.627	7.400	4.644	-0.774	0.356	-1.582	-4.357
3	6.942	6.821	4.560	6.189	4.745	2.998	0.173	0.603	-2.603	-1.687
4	-0.311	0.235	0.510	4.302	3.213	7.507	3.164	-1.248	-2.639	-3.312
Big	-3.008	-2.463	-3.736	-1.342	2.564	7.825	4.197	3.486	-1.799	-5.151

t(p)						t(t)				
	Low	2	3	4	High	Low	2	3	4	High
Small	0.073	-1.639	-1.797	-1.340	-2.285	-2.697	-1.961	-1.187	-2.297	-0.054
2	-0.244	-3.559	-1.931	-2.928	-3.483	0.509	-1.496	-1.552	-2.775	-2.398
3	-3.946	-3.560	-1.282	-2.527	-0.568	-0.443	-1.651	0.740	-3.265	-2.223
4	-1.578	-1.806	-4.412	-2.968	-0.997	-2.794	-1.876	-3.476	-3.705	-2.350
Big	-1.637	-3.059	-0.027	-1.726	-2.139	-1.586	-1.539	0.837	-0.793	-1.542

Panel B MC_ROE

a						b				
	Low	2	3	4	High	Low	2	3	4	High
Small	0.011	0.006	0.006	0.005	0.004	0.995	0.963	0.941	1.025	1.112
2	0.001	0.002	0.001	-0.001	0.002	0.957	0.994	1.005	1.014	1.044
3	-0.003	0.005	0.000	-0.004	0.006	0.994	1.043	1.093	0.963	1.010
4	0.004	0.002	0.008	0.008	0.010	1.021	0.926	0.952	0.966	0.916
Big	0.002	-0.002	0.003	0.003	0.004	1.108	1.157	0.973	1.014	1.019

s						l				
	Low	2	3	4	High	Low	2	3	4	High
Small	1.744	1.185	1.320	1.469	1.608	0.278	0.076	0.333	0.425	0.068
2	1.092	0.877	0.704	1.107	0.762	0.249	0.080	0.000	-0.331	0.033
3	0.893	0.328	0.489	0.595	0.381	-0.037	-0.188	0.303	0.213	0.032
4	0.330	0.201	-0.059	-0.129	0.293	-0.256	0.355	0.194	0.217	-0.259
Big	-0.196	-0.213	-0.071	0.131	0.026	0.024	0.241	-0.058	-0.152	0.011

p						t				
	Low	2	3	4	High	Low	2	3	4	High
Small	-0.405	-0.562	-0.211	0.281	0.326	0.036	-0.363	-0.335	-0.499	-0.189
2	-0.849	-0.630	-0.145	-0.260	0.163	-0.223	-0.378	-0.028	-0.355	0.027
3	-1.084	-0.903	-0.032	-0.023	0.024	-0.359	-0.301	-0.370	-0.429	-0.288
4	-1.263	-0.667	-0.339	0.057	0.010	-0.365	-0.665	-0.263	0.038	-0.118
Big	-0.731	-0.324	-0.432	-0.183	0.228	-0.004	0.017	-0.127	-0.234	-0.245

t(a)						t(b)				
	Low	2	3	4	High	Low	2	3	4	High
Small	2.534	1.366	1.448	1.406	0.961	15.527	15.973	15.108	18.812	18.672
2	0.332	0.592	0.257	-0.229	0.446	16.836	18.462	20.417	18.783	20.343
3	-0.635	1.290	-0.012	-1.042	1.900	16.261	18.573	20.542	19.205	21.680
4	0.966	0.601	2.252	1.977	3.370	18.141	16.411	17.458	16.312	21.590
Big	0.676	-0.767	1.261	0.882	1.229	23.416	26.259	24.953	24.583	22.882

t(s)						t(l)				
	Low	2	3	4	High	Low	2	3	4	High
Small	12.490	9.020	9.729	12.372	12.382	1.677	0.487	2.069	3.018	0.444
2	8.818	7.477	6.557	9.412	6.811	1.692	0.576	0.002	-2.369	0.249
3	6.703	2.678	4.218	5.443	3.751	-0.232	-1.292	2.200	1.645	0.268
4	2.687	1.631	-0.498	-0.996	3.167	-1.758	2.435	1.376	1.415	-2.359
Big	-1.904	-2.223	-0.841	1.456	0.265	0.194	2.112	-0.579	-1.423	0.096

t(p)						t(t)				
	Low	2	3	4	High	Low	2	3	4	High
Small	-2.127	-3.134	-1.139	1.733	1.843	0.207	-2.217	-1.979	-3.371	-1.172
2	-5.026	-3.936	-0.990	-1.622	1.068	-1.445	-2.588	-0.207	-2.419	0.195
3	-5.964	-5.410	-0.203	-0.151	0.174	-2.165	-1.972	-2.562	-3.150	-2.275
4	-7.553	-3.980	-2.093	0.325	0.076	-2.392	-4.343	-1.776	0.234	-1.023
Big	-5.198	-2.475	-3.726	-1.495	1.721	-0.030	0.142	-1.198	-2.089	-2.023

Panel C MC_TA

a						b				
	Low	2	3	4	High	Low	2	3	4	High
Small	0.010	0.008	0.005	0.005	0.009	1.071	1.085	0.993	0.960	1.010
2	0.002	0.002	0.000	0.001	0.003	1.060	1.030	0.935	0.914	0.942
3	0.001	-0.003	-0.003	0.001	-0.004	1.057	0.984	1.113	1.065	0.920
4	0.005	0.009	0.005	0.001	0.005	1.078	0.932	1.025	0.986	1.243
Big	0.005	0.003	0.000	0.008	0.004	0.942	0.829	1.042	1.057	1.115

s						l				
	Low	2	3	4	High	Low	2	3	4	High
Small	1.687	1.546	1.219	1.340	1.699	0.159	0.232	0.202	0.176	0.226
2	0.872	0.895	0.865	0.641	1.086	0.322	-0.040	0.088	0.235	-0.166
3	0.662	0.352	0.480	0.418	0.843	0.098	0.202	0.129	-0.004	0.023
4	0.152	0.122	0.101	0.086	0.263	-0.178	-0.036	0.297	0.262	-0.207
Big	0.068	-0.150	-0.122	-0.269	0.148	-0.272	0.159	0.017	0.265	-0.361

p						t				
	Low	2	3	4	High	Low	2	3	4	High
Small	-0.253	-0.385	-0.318	-0.156	-0.144	0.417	-0.054	-0.250	-0.563	-0.813
2	-0.285	-0.501	-0.445	-0.249	-0.691	-0.053	-0.117	-0.147	-0.267	-0.881
3	-0.420	-0.103	-0.176	-0.389	-0.589	0.272	0.241	-0.350	-0.518	-1.179
4	-0.446	-0.587	-0.071	-0.132	-0.620	0.063	-0.285	0.084	-0.220	-1.000
Big	-0.476	-0.128	-0.067	0.108	-0.670	0.351	0.194	0.075	-0.203	-0.838

t(a)						t(b)				
	Low	2	3	4	High	Low	2	3	4	High
Small	2.365	2.032	1.266	1.519	2.049	18.194	18.460	17.669	15.022	15.801
2	0.483	0.403	-0.032	0.314	0.694	19.064	18.067	19.344	17.497	15.039
3	0.310	-0.714	-0.818	0.300	-0.826	22.709	18.422	22.994	20.072	14.165
4	1.231	2.536	1.388	0.325	1.863	17.672	19.144	21.138	21.846	19.704
Big	1.643	0.862	0.084	3.039	1.398	22.996	19.459	31.154	28.537	26.464

t(s)						t(l)				
	Low	2	3	4	High	Low	2	3	4	High
Small	13.148	12.063	9.953	9.620	12.190	1.042	1.524	1.391	1.063	1.365
2	7.200	7.205	8.213	5.624	7.950	2.239	-0.269	0.706	1.741	-1.025
3	6.529	3.020	4.556	3.614	5.961	0.813	1.462	1.030	-0.027	0.135
4	1.144	1.146	0.958	0.870	1.910	-1.126	-0.284	2.368	2.241	-1.269
Big	0.765	-1.617	-1.667	-3.333	1.609	-2.564	1.444	0.191	2.770	-3.318

t(p)						t(t)				
	Low	2	3	4	High	Low	2	3	4	High
Small	-1.447	-2.205	-1.902	-0.819	-0.760	2.611	-0.339	-1.639	-3.245	-4.684
2	-1.728	-2.958	-3.099	-1.605	-3.714	-0.350	-0.753	-1.122	-1.883	-5.180
3	-3.037	-0.648	-1.226	-2.468	-3.053	2.151	1.660	-2.662	-3.591	-6.688
4	-2.458	-4.059	-0.491	-0.980	-3.310	0.380	-2.155	0.637	-1.792	-5.837
Big	-3.911	-1.015	-0.669	0.980	-5.351	3.158	1.675	0.826	-2.023	-7.330

Panel D MC_HC

a						b				
	Low	2	3	4	High	Low	2	3	4	High
Small	0.000	0.016	0.003	0.001	0.008	1.243	0.868	0.943	1.026	0.918
2	0.006	0.003	-0.005	0.001	0.002	1.075	1.015	0.980	0.885	0.873
3	-0.003	0.001	0.003	-0.002	-0.002	1.087	0.981	1.043	1.092	0.993
4	0.006	0.003	0.008	0.005	0.002	1.082	0.801	0.867	0.973	0.953
Big	0.006	0.006	0.004	0.005	-0.002	0.937	1.261	1.041	0.964	1.051

s						l				
	Low	2	3	4	High	Low	2	3	4	High
Small	1.516	1.175	1.271	1.479	1.411	0.478	0.533	0.209	0.177	-0.001
2	1.046	0.951	0.772	0.907	0.607	-0.196	0.032	0.404	0.139	0.214
3	0.771	0.565	0.367	0.652	0.342	-0.028	-0.099	0.203	-0.282	0.363
4	0.148	0.098	0.259	0.291	-0.049	-0.080	0.240	0.312	-0.002	0.402
Big	-0.341	0.089	0.179	-0.028	0.063	-0.039	-0.209	-0.156	-0.005	-0.095

p						t				
	Low	2	3	4	High	Low	2	3	4	High
Small	-0.175	-0.227	-0.206	0.151	0.019	-0.128	-0.315	-0.056	0.013	-0.221
2	-0.900	-0.588	-0.023	-0.184	-0.243	-0.514	-0.193	-0.108	0.106	-0.298
3	-0.680	-0.539	-0.176	-0.520	0.098	-0.245	-0.093	-0.439	-0.557	-0.141
4	-0.829	-0.144	-0.237	-0.193	-0.016	-0.176	0.079	-0.623	-0.284	0.157
Big	-0.547	-0.398	-0.168	-0.046	-0.098	-0.003	-0.333	-0.216	-0.121	-0.064

t(a)						t(b)				
	Low	2	3	4	High	Low	2	3	4	High
Small	-0.087	3.338	0.820	0.172	1.985	19.470	12.170	15.782	15.849	14.107
2	1.464	0.847	-1.250	0.279	0.446	16.963	20.394	18.347	16.522	14.903
3	-0.650	0.333	0.940	-0.551	-0.562	17.381	18.492	19.529	19.588	21.744
4	1.608	0.951	1.983	1.398	0.445	17.670	16.138	14.578	18.803	15.482
Big	1.426	1.426	1.237	1.789	-0.761	13.836	21.532	22.190	25.198	28.878

t(s)						t(l)				
	Low	2	3	4	High	Low	2	3	4	High
Small	10.900	7.553	9.756	10.484	9.943	2.897	2.886	1.350	1.057	-0.003
2	7.569	8.763	6.632	7.762	4.756	-1.197	0.251	2.922	1.003	1.414
3	5.654	4.889	3.156	5.364	3.432	-0.173	-0.722	1.472	-1.954	3.076
4	1.109	0.907	1.995	2.581	-0.363	-0.507	1.871	2.029	-0.018	2.526
Big	-2.310	0.696	1.751	-0.335	0.795	-0.222	-1.381	-1.289	-0.049	-1.011

t(p)						t(t)				
	Low	2	3	4	High	Low	2	3	4	High
Small	-0.921	-1.072	-1.157	0.787	0.097	-0.741	-1.625	-0.347	0.073	-1.249
2	-4.775	-3.977	-0.144	-1.155	-1.398	-2.984	-1.427	-0.745	0.729	-1.873
3	-3.659	-3.421	-1.112	-3.135	0.724	-1.442	-0.644	-3.026	-3.679	-1.134
4	-4.558	-0.976	-1.338	-1.258	-0.088	-1.060	0.583	-3.859	-2.021	0.942
Big	-2.719	-2.289	-1.205	-0.402	-0.905	-0.016	-2.096	-1.700	-1.165	-0.644

Panel E R²

		MC_PB				MC_ROE					
		Low	2	3	4	High	Low	2	3	4	High
Small		0.779	0.852	0.839	0.835	0.807	0.819	0.803	0.782	0.815	0.783
	2	0.828	0.776	0.811	0.832	0.803	0.859	0.843	0.817	0.786	0.792
	3	0.884	0.863	0.785	0.851	0.819	0.839	0.841	0.832	0.809	0.825
	4	0.885	0.802	0.861	0.830	0.791	0.861	0.841	0.811	0.746	0.804
	Big	0.897	0.915	0.917	0.874	0.871	0.895	0.903	0.889	0.863	0.831

		MC_TA				MC_HC					
		Low	2	3	4	High	Low	2	3	4	High
Small		0.838	0.844	0.821	0.755	0.789	0.848	0.735	0.782	0.746	0.699
	2	0.838	0.813	0.844	0.807	0.775	0.817	0.861	0.813	0.780	0.751
	3	0.875	0.799	0.859	0.834	0.766	0.813	0.817	0.824	0.817	0.843
	4	0.783	0.840	0.843	0.857	0.836	0.827	0.772	0.759	0.797	0.754
	Big	0.860	0.824	0.911	0.903	0.901	0.737	0.839	0.834	0.869	0.894

Table 6: Four factor regression to explain the average returns on the fifth

	Intercepts				
	Rm	SMB	LMH	RMW	CMA
c	0.019	0.012	0.008	0.007	0.001
rm		-0.090	-0.018	-0.111	-0.100
s	-0.394		0.074	0.076	0.216
l	-0.115	0.108		-0.692	-0.091
p	-0.978	0.152	-0.940		-0.388
t	-0.733	0.362	-0.104	-0.326	
	T-Statistics				
c	3.336	4.550	3.617	4.042	0.985
rm	-2.295	-2.295	-0.542	-4.197	-3.371
s	-0.542	1.078	1.078	1.302	3.494
l	-4.197	1.302	-16.376	-16.376	-1.173
p	-3.371	3.494	-1.173	-4.563	-4.563
t			R2		
	0.310	0.160	0.733	0.781	0.299

Table 7 Six factor regression results for 25 portfolios, MC_ROE, MC_TA and MC_HC cross portfolios.

$$R_{Pt} - R_{Ft} = a + b (R_{Mt} - R_{Ft}) + s \text{ SMB}_t + l \text{ LMH}_t + p \text{ RMW}_t + t \text{ CMA}_t + h \text{ LcMH}_t + e_t$$

Panel A MC_PB

a						RM				
	Low	2	3	4	High	Low	2	3	4	High
Small	0.014	0.010	0.012	0.003	0.000	0.854	1.091	1.099	1.003	1.060
2	0.000	0.004	0.002	-0.003	0.005	0.947	0.976	0.879	0.942	0.946
3	0.000	-0.001	-0.003	0.001	0.003	1.010	1.137	0.916	1.094	0.973
4	-0.002	0.002	0.008	0.003	0.005	1.019	0.940	1.174	1.038	0.840
Big	0.006	0.000	0.002	0.006	0.003	0.964	1.072	1.039	1.087	0.899

S						L				
	Low	2	3	4	High	Low	2	3	4	High
Small	1.586	1.560	1.214	1.477	1.300	0.732	0.196	-0.071	-0.194	-0.583
2	1.167	0.941	0.944	1.106	0.772	0.710	-0.107	0.047	-0.205	-0.544
3	0.759	0.788	0.521	0.637	0.459	0.389	0.023	0.073	-0.319	-0.196
4	-0.023	0.039	0.036	0.458	0.255	1.056	0.490	-0.139	-0.311	-0.327
Big	-0.330	-0.268	-0.308	-0.122	0.198	0.978	0.492	0.335	-0.180	-0.453

R						T				
	Low	2	3	4	High	Low	2	3	4	High
Small	0.316	-0.241	-0.219	-0.210	-0.351	-0.458	-0.286	-0.168	-0.307	-0.001
2	0.067	-0.511	-0.178	-0.360	-0.429	0.103	-0.240	-0.196	-0.348	-0.302
3	-0.503	-0.497	-0.186	-0.303	-0.041	-0.051	-0.228	0.108	-0.410	-0.264
4	-0.262	-0.343	-0.548	-0.249	0.059	-0.419	-0.312	-0.514	-0.490	-0.248
Big	-0.273	-0.279	0.056	-0.128	-0.226	-0.208	-0.160	0.087	-0.090	-0.142

	H					t(h)				
	Low	2	3	4	High	Low	2	3	4	High
Small	0.521	0.028	0.099	-0.046	-0.011	3.296	0.205	0.774	-0.391	-0.084
2	0.174	0.300	0.174	0.079	0.116	1.187	1.879	1.444	0.669	0.963
3	0.138	0.106	0.015	0.082	0.055	1.124	0.804	0.113	0.701	0.494
4	-0.008	-0.030	0.343	0.348	0.326	-0.059	-0.203	2.518	2.867	3.152
Big	-0.067	0.186	0.106	0.172	-0.026	-0.563	1.806	1.248	1.630	-0.316

	t(a)					t(RM)				
	Low	2	3	4	High	Low	2	3	4	High
Small	3.086	2.482	3.384	0.845	0.043	13.039	19.137	20.717	20.776	19.522
2	0.031	0.882	0.656	-0.891	1.615	15.592	14.776	17.663	19.332	18.994
3	0.106	-0.255	-0.951	0.217	0.894	19.849	20.788	17.161	22.628	21.147
4	-0.427	0.579	1.964	0.919	1.581	18.234	15.161	20.785	20.667	19.634
Big	1.682	-0.121	0.698	1.802	1.210	19.578	25.181	29.446	24.884	26.373

	t(s)					t(l)				
	Low	2	3	4	High	Low	2	3	4	High
Small	11.003	12.433	10.400	13.899	10.879	4.357	1.339	-0.524	-1.563	-4.187
2	8.729	6.481	8.623	10.313	7.045	4.555	-0.634	0.370	-1.642	-4.256
3	6.780	6.548	4.440	5.990	4.535	2.979	0.167	0.536	-2.575	-1.662
4	-0.184	0.285	0.291	4.147	2.703	7.369	3.081	-0.961	-2.413	-2.977
Big	-3.048	-2.866	-3.962	-1.270	2.636	7.745	4.508	3.706	-1.606	-5.180

t(r)						t(T)					
	Low	2	3	4	High	Low	2	3	4	High	
Small	1.502	-1.317	-1.289	-1.355	-2.015	-2.622	-1.880	-1.189	-2.388	-0.005	
2	0.343	-2.409	-1.114	-2.298	-2.685	0.637	-1.362	-1.474	-2.678	-2.278	
3	-3.080	-2.834	-1.085	-1.956	-0.275	-0.375	-1.562	0.761	-3.187	-2.150	
4	-1.464	-1.721	-3.021	-1.545	0.433	-2.816	-1.886	-3.416	-3.662	-2.176	
Big	-1.726	-2.042	0.492	-0.913	-2.064	-1.586	-1.410	0.927	-0.776	-1.567	

Panel B MC_ROE

a						RM					
	Low	2	3	4	High	Low	2	3	4	High	
Small	0.011	0.005	0.006	0.005	0.004	0.977	0.939	0.925	1.016	1.114	
2	0.001	0.002	0.001	-0.001	0.002	0.950	0.983	0.999	1.003	1.045	
3	-0.004	0.005	0.000	-0.004	0.005	0.958	1.049	1.082	0.955	0.983	
4	0.004	0.002	0.008	0.006	0.009	1.020	0.918	0.946	0.959	0.900	
Big	0.002	-0.002	0.003	0.002	0.003	1.090	1.156	0.975	0.996	1.010	

S						L					
	Low	2	3	4	High	Low	2	3	4	High	
Small	1.770	1.180	1.344	1.489	1.657	0.263	0.087	0.325	0.406	0.027	
2	1.120	0.886	0.698	1.112	0.769	0.233	0.076	-0.005	-0.339	0.016	
3	0.857	0.348	0.495	0.597	0.352	0.002	-0.206	0.307	0.211	0.068	
4	0.325	0.203	-0.065	-0.142	0.273	-0.261	0.362	0.205	0.231	-0.239	
Big	-0.244	-0.215	-0.074	0.099	0.034	0.059	0.247	-0.059	-0.123	0.022	

R					T					
	Low	2	3	4	High	Low	2	3	4	High
Small	-0.290	-0.402	-0.105	0.337	0.316	0.048	-0.344	-0.326	-0.490	-0.192
2	-0.798	-0.561	-0.110	-0.196	0.152	-0.220	-0.370	-0.018	-0.344	0.031
3	-0.850	-0.942	0.041	0.025	0.205	-0.329	-0.306	-0.365	-0.422	-0.267
4	-1.262	-0.611	-0.295	0.102	0.114	-0.361	-0.662	-0.260	0.043	-0.104
Big	-0.621	-0.314	-0.445	-0.063	0.295	0.015	0.016	-0.127	-0.217	-0.244

H					t(h)					
	Low	2	3	4	High	Low	2	3	4	High
Small	0.185	0.269	0.176	0.082	-0.036	1.178	1.823	1.154	0.609	-0.247
2	0.080	0.112	0.050	0.099	-0.031	0.569	0.840	0.409	0.737	-0.244
3	0.404	-0.073	0.126	0.077	0.317	2.728	-0.524	0.959	0.619	2.820
4	-0.004	0.099	0.081	0.082	0.181	-0.025	0.710	0.600	0.559	1.739
Big	0.193	0.021	-0.024	0.211	0.124	1.674	0.193	-0.252	2.096	1.133

t(a)					t(RM)					
	Low	2	3	4	High	Low	2	3	4	High
Small	2.426	1.215	1.328	1.370	0.989	14.992	15.328	14.646	18.135	18.347
2	0.265	0.522	0.276	-0.251	0.519	16.413	17.871	19.687	18.071	19.770
3	-0.862	1.326	-0.120	-1.074	1.678	15.604	18.280	19.969	18.614	21.131
4	0.999	0.508	2.165	1.901	3.230	17.668	15.933	16.959	15.842	20.895
Big	0.581	-0.803	1.289	0.728	1.043	22.808	25.608	24.433	23.896	22.268

t(s)					t(l)					
	Low	2	3	4	High	Low	2	3	4	High
Small	12.345	8.760	9.674	12.085	12.404	1.576	0.556	2.008	2.824	0.171
2	8.802	7.316	6.254	9.103	6.614	1.571	0.539	-0.035	-2.382	0.115
3	6.343	2.758	4.154	5.292	3.435	0.010	-1.399	2.208	1.605	0.568
4	2.557	1.605	-0.531	-1.068	2.884	-1.760	2.452	1.436	1.488	-2.166
Big	-2.320	-2.165	-0.846	1.083	0.343	0.485	2.132	-0.579	-1.155	0.192

t(r)					t(T)					
	Low	2	3	4	High	Low	2	3	4	High
Small	-1.386	-2.046	-0.517	1.877	1.622	0.276	-2.106	-1.937	-3.283	-1.187
2	-4.300	-3.178	-0.675	-1.100	0.898	-1.429	-2.526	-0.135	-2.323	0.220
3	-4.309	-5.115	0.233	0.151	1.371	-2.011	-2.003	-2.524	-3.089	-2.151
4	-6.809	-3.307	-1.647	0.527	0.823	-2.347	-4.312	-1.747	0.267	-0.908
Big	-4.047	-2.170	-3.478	-0.469	2.025	0.117	0.135	-1.196	-1.955	-2.018

Panel C MC_TA

a					RM					
	Low	2	3	4	High	Low	2	3	4	High
Small	0.006	0.006	0.005	0.006	0.006	1.063	1.061	0.980	0.948	0.991
2	0.002	0.002	-0.001	0.001	0.003	1.050	1.027	0.915	0.916	0.942
3	0.001	-0.003	-0.003	0.001	-0.003	1.048	0.970	1.106	1.049	0.927
4	0.004	0.008	0.005	0.001	0.005	1.056	0.923	1.021	0.986	1.212
Big	0.004	0.003	0.000	0.006	0.004	0.926	0.842	1.035	1.035	1.106

S						L					
	Low	2	3	4	High	Low	2	3	4	High	
Small	1.676	1.515	1.203	1.324	1.674	0.172	0.270	0.223	0.195	0.256	
2	0.860	0.892	0.839	0.643	1.086	0.337	-0.036	0.120	0.233	-0.166	
3	0.651	0.334	0.472	0.397	0.853	0.111	0.224	0.139	0.022	0.011	
4	0.124	0.110	0.096	0.086	0.223	-0.144	-0.022	0.304	0.261	-0.159	
Big	0.048	-0.133	-0.131	-0.297	0.136	-0.246	0.138	0.028	0.299	-0.347	

R						T					
	Low	2	3	4	High	Low	2	3	4	High	
Small	-0.195	-0.224	-0.232	-0.075	-0.017	0.424	-0.036	-0.241	-0.554	-0.799	
2	-0.221	-0.485	-0.312	-0.260	-0.692	-0.046	-0.115	-0.133	-0.268	-0.881	
3	-0.363	-0.009	-0.133	-0.281	-0.640	0.278	0.251	-0.345	-0.506	-1.185	
4	-0.303	-0.527	-0.043	-0.134	-0.418	0.079	-0.278	0.087	-0.220	-0.977	
Big	-0.369	-0.219	-0.017	0.251	-0.611	0.363	0.184	0.081	-0.188	-0.832	

H						t(h)					
	Low	2	3	4	High	Low	2	3	4	High	
Small	0.103	0.287	0.152	0.144	0.227	0.709	2.000	1.098	0.912	1.446	
2	0.115	0.028	0.236	-0.020	-0.001	0.840	0.201	2.002	-0.151	-0.008	
3	0.101	0.167	0.077	0.193	-0.090	0.882	1.270	0.644	1.481	-0.561	
4	0.254	0.107	0.050	-0.004	0.360	1.698	0.890	0.416	-0.039	2.351	
Big	0.191	-0.160	0.089	0.255	0.105	1.910	-1.532	1.077	2.860	1.009	

		t(a)					t(RM)				
		Low	2	3	4	High	Low	2	3	4	High
Small		0.058	1.849	1.155	1.422	1.907	17.640	17.857	17.090	14.515	15.236
	2	0.399	0.380	-0.226	0.326	0.689	18.477	17.590	18.732	17.104	14.676
	3	0.223	-0.835	-0.875	0.156	-0.765	22.039	17.818	22.338	19.434	13.951
	4	1.069	2.436	1.337	0.326	1.656	17.073	18.550	20.553	21.324	19.119
Big		1.465	1.011	-0.021	1.822	1.294	22.336	19.454	30.303	28.054	25.709

		t(s)					t(l)				
		Low	2	3	4	High	Low	2	3	4	High
Small		12.940	11.853	9.751	9.431	11.969	1.121	1.781	1.520	1.169	1.543
	2	7.037	7.103	7.993	5.581	7.864	2.325	-0.240	0.961	1.702	-1.014
	3	6.367	2.849	4.434	3.421	5.970	0.918	1.614	1.101	0.160	0.063
	4	0.935	1.028	0.897	0.865	1.638	-0.910	-0.169	2.395	2.210	-0.982
Big		0.533	-1.426	-1.787	-3.739	1.474	-2.325	1.246	0.326	3.177	-3.164

		t(r)					t(T)				
		Low	2	3	4	High	Low	2	3	4	High
Small		-1.009	-1.174	-1.261	-0.356	-0.079	2.642	-0.229	-1.576	-3.186	-4.612
	2	-1.209	-2.586	-1.992	-1.513	-3.357	-0.302	-0.738	-1.019	-1.882	-5.154
	3	-2.377	-0.051	-0.837	-1.620	-2.997	2.196	1.732	-2.616	-3.517	-6.693
	4	-1.525	-3.299	-0.268	-0.902	-2.052	0.478	-2.100	0.658	-1.785	-5.786
Big		-2.770	-1.572	-0.152	2.119	-4.423	3.290	1.594	0.886	-1.908	-7.261

Panel D MC_HC

a						RM				
	Low	2	3	4	High	Low	2	3	4	High
Small	-0.002	0.014	0.004	0.002	0.005	1.207	0.804	0.968	1.053	0.991
2	0.004	0.002	-0.005	0.001	0.003	1.024	0.986	0.965	0.885	0.904
3	-0.004	0.001	0.003	-0.002	-0.001	1.044	0.978	1.036	1.109	1.014
4	0.006	0.003	0.007	0.006	0.003	1.035	0.786	0.848	1.020	0.988
Big	0.005	0.004	0.003	0.005	-0.001	0.893	1.208	1.020	0.981	1.070

S						L				
	Low	2	3	4	High	Low	2	3	4	High
Small	1.524	1.126	1.343	1.557	1.561	0.498	0.608	0.138	0.097	-0.154
2	1.051	0.951	0.780	0.924	0.673	-0.144	0.055	0.408	0.118	0.148
3	0.701	0.555	0.369	0.707	0.378	0.028	-0.105	0.205	-0.328	0.322
4	0.056	0.079	0.239	0.361	-0.026	-0.004	0.262	0.338	-0.086	0.350
Big	-0.410	0.024	0.151	-0.017	0.072	0.041	-0.126	-0.127	-0.031	-0.126

R						T				
	Low	2	3	4	High	Low	2	3	4	High
Small	0.070	0.207	-0.367	-0.033	-0.461	-0.106	-0.269	-0.076	-0.008	-0.280
2	-0.537	-0.390	0.077	-0.187	-0.450	-0.489	-0.176	-0.099	0.108	-0.324
3	-0.406	-0.529	-0.134	-0.630	-0.041	-0.203	-0.084	-0.434	-0.574	-0.156
4	-0.532	-0.045	-0.106	-0.508	-0.257	-0.131	0.090	-0.609	-0.319	0.138
Big	-0.252	-0.044	-0.034	-0.169	-0.234	0.030	-0.295	-0.200	-0.131	-0.072

	H					t(h)				
	Low	2	3	4	High	Low	2	3	4	High
Small	0.422	0.761	-0.301	-0.344	-0.866	2.763	4.626	-2.081	-2.199	-6.164
2	0.645	0.346	0.171	-0.017	-0.373	4.472	2.911	1.307	-0.129	-2.651
3	0.472	0.007	0.072	-0.202	-0.252	3.121	0.055	0.543	-1.483	-2.273
4	0.520	0.174	0.229	-0.565	-0.433	3.582	1.429	1.571	-4.762	-2.934
Big	0.529	0.631	0.236	-0.222	-0.248	3.284	4.683	2.064	-2.401	-2.832

	t(a)					t(RM)				
	Low	2	3	4	High	Low	2	3	4	High
Small	-0.379	3.096	1.031	0.399	1.513	19.076	11.785	16.157	16.235	17.013
2	1.026	0.554	-1.388	0.321	0.696	17.124	20.044	17.797	16.092	15.509
3	-0.864	0.396	0.885	-0.447	-0.341	16.657	17.890	18.977	19.709	22.095
4	1.412	0.819	1.737	1.863	0.799	17.210	15.578	14.032	20.762	16.148
Big	1.150	1.057	1.066	1.896	-0.414	13.374	21.648	21.542	25.579	29.497

	t(s)					t(l)				
	Low	2	3	4	High	Low	2	3	4	High
Small	10.951	7.507	10.195	10.908	12.186	3.069	3.480	0.897	0.580	-1.034
2	7.994	8.792	6.538	7.635	5.248	-0.938	0.432	2.933	0.838	0.988
3	5.085	4.613	3.071	5.710	3.745	0.173	-0.750	1.461	-2.270	2.733
4	0.420	0.713	1.796	3.336	-0.193	-0.025	2.026	2.181	-0.680	2.231
Big	-2.790	0.199	1.450	-0.206	0.899	0.240	-0.881	-1.044	-0.318	-1.349

t(r)						t(T)					
	Low	2	3	4	High	Low	2	3	4	High	
Small	0.344	0.945	-1.909	-0.156	-2.465	-0.631	-1.479	-0.478	-0.047	-1.802	
2	-2.799	-2.470	0.444	-1.058	-2.405	-3.072	-1.340	-0.687	0.738	-2.084	
3	-2.017	-3.013	-0.764	-3.489	-0.277	-1.216	-0.577	-2.983	-3.830	-1.276	
4	-2.754	-0.280	-0.548	-3.225	-1.307	-0.819	0.670	-3.781	-2.434	0.848	
Big	-1.177	-0.247	-0.221	-1.372	-2.010	0.170	-1.985	-1.582	-1.276	-0.748	

Panel E R^2

MCXPB_R2						MCXROE_R2					
	Low	2	3	4	High	Low	2	3	4	High	
Small	0.796	0.849	0.843	0.841	0.802	0.822	0.806	0.786	0.814	0.816	
2	0.824	0.838	0.824	0.830	0.802	0.861	0.843	0.815	0.844	0.790	
3	0.885	0.862	0.804	0.850	0.818	0.845	0.841	0.834	0.809	0.833	
4	0.885	0.802	0.867	0.840	0.803	0.860	0.842	0.812	0.797	0.807	
Big	0.898	0.917	0.918	0.876	0.872	0.897	0.903	0.889	0.866	0.833	

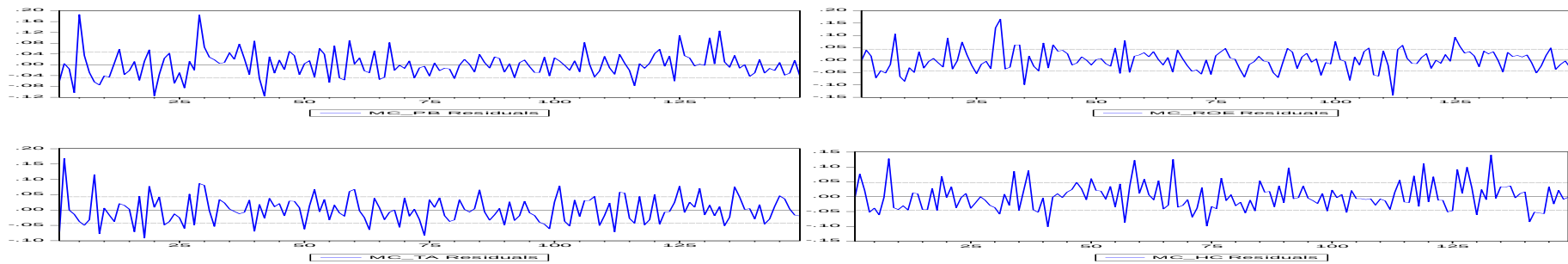
MCXTA_R2						MCXHC_R2					
	Low	2	3	4	High	Low	2	3	4	High	
Small	0.854	0.849	0.822	0.827	0.792	0.858	0.769	0.791	0.756	0.770	
2	0.839	0.813	0.848	0.807	0.775	0.844	0.870	0.816	0.779	0.765	
3	0.875	0.801	0.860	0.836	0.767	0.821	0.814	0.824	0.822	0.849	
4	0.787	0.841	0.844	0.857	0.842	0.841	0.776	0.763	0.826	0.768	
Big	0.864	0.827	0.912	0.908	0.901	0.756	0.860	0.839	0.874	0.899	

Table 8 Summary of the Factor regressions

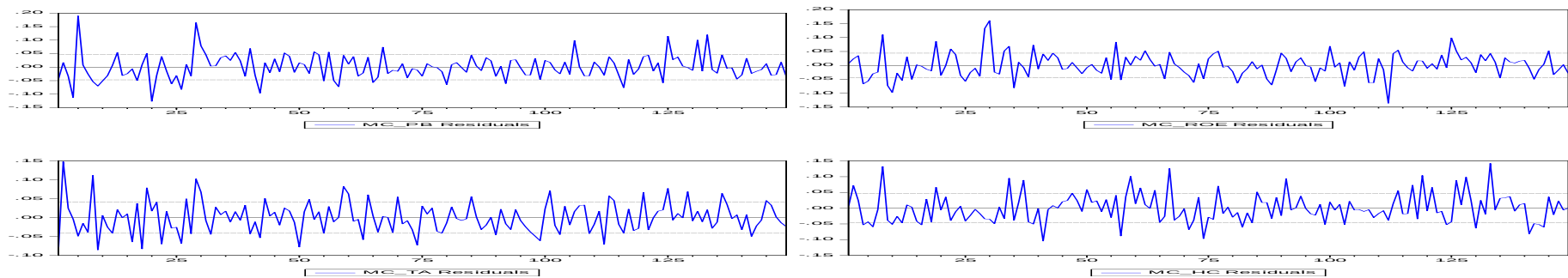
Model	No of significant intercepts				R-Square			
	MC_PB	MC_ROE	MC_TA	MC_HC	MC_PB	MC_ROE	MC_TA	MC_HC
3 Factor	4	6	5	3	82%	81%	81%	79%
5 Factor	4	4	5	3	84%	83%	83%	80%
6 Factor	3	3	2	1	85%	84%	84%	81%

Figure 3: Residual graphs for three, five and six factor regression for Portfolio 1

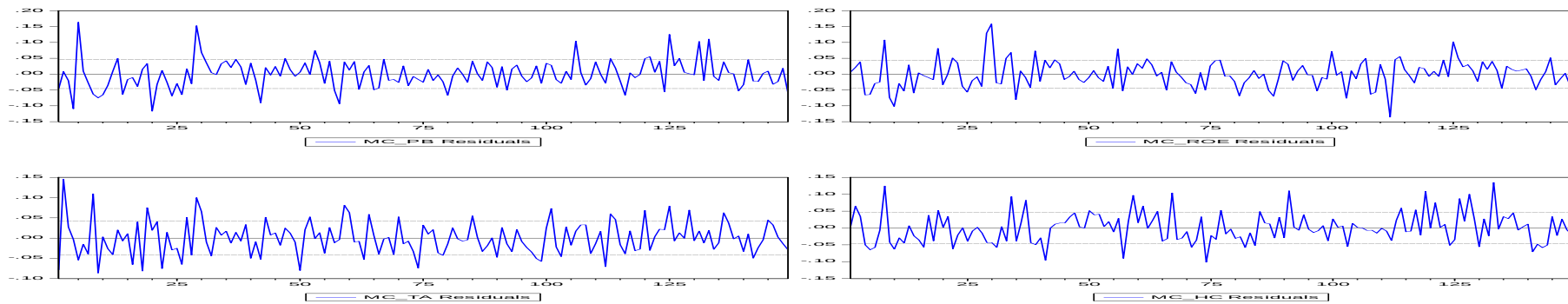
Three factor (MC_PB, MC_ROE, MC_TA and MC_HC)



Five factor (MC_PB, MC_ROE, MC_TA and MC_HC)

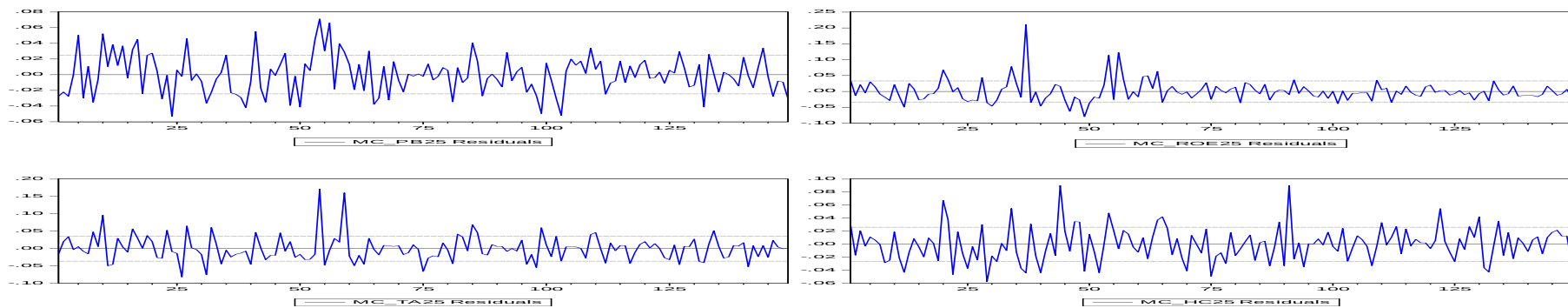


Six factor (MC_PB, MC_ROE, MC_TA and MC_HC)

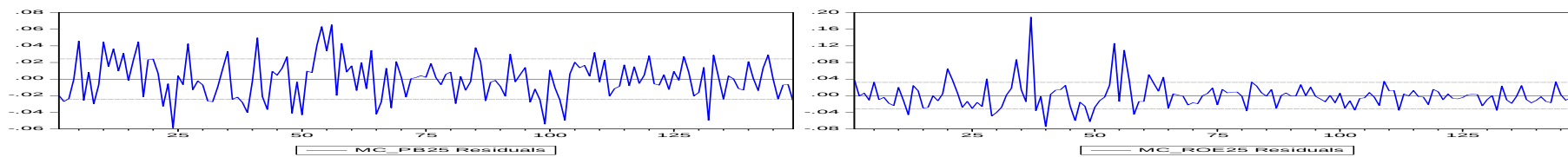


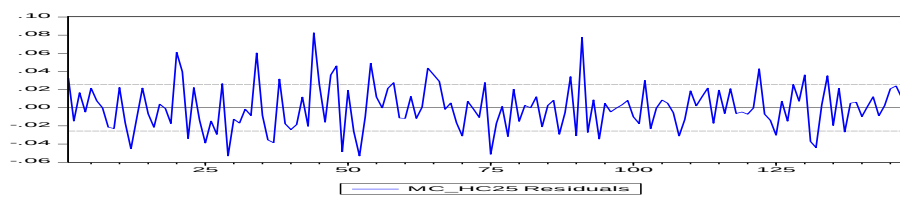
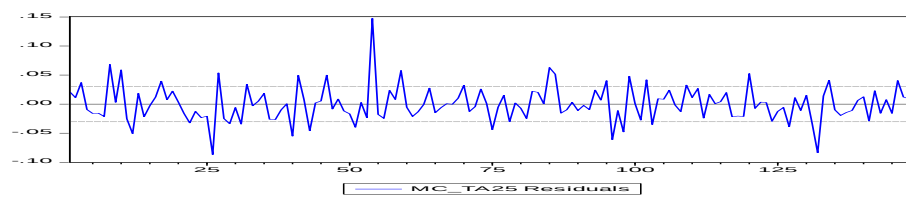
Residual graphs for three, five and six factor regression for Portfolio 25

Three factor (MC_PB, MC_ROE, MC_TA and MC_HC)



Five factor (MC_PB, MC_ROE, MC_TA and MC_HC)





Six factor (MC_PB, MC_ROE, MC_TA and MC_HC)

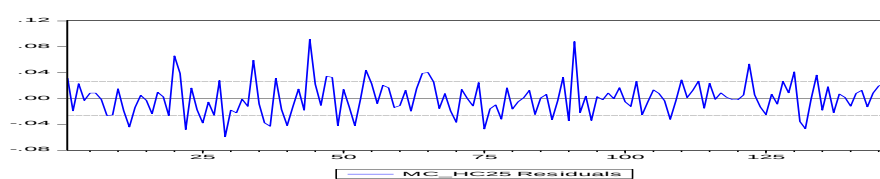
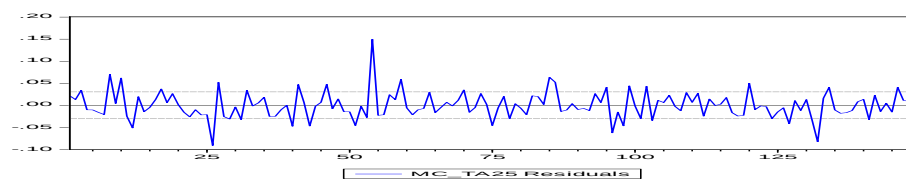
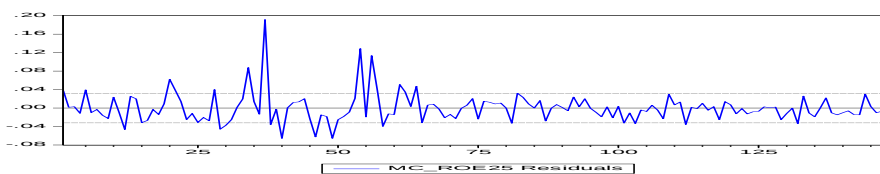
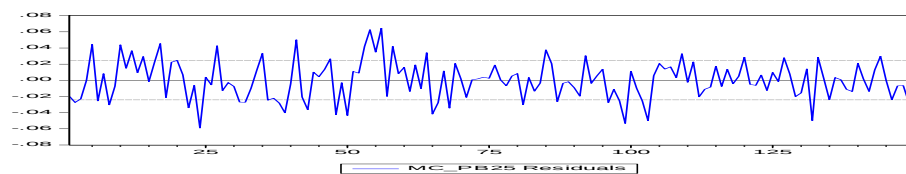


Table 9 Summary of GRS test results for all the Factor models

Three factor	GRS F-Statistics	P-Value	Average absolute alpha value	Average R2
MC_PB*	1.746	0.025	0.004	0.82
MC_ROE*	2.086	0.004	0.005	0.81
MC_TA*	1.596	0.050	0.004	0.81
MC_HC*	1.598	0.050	0.004	0.79
Five factor	GRS F-Statistics	P-Value	Average absolute alpha value	Average R2
MC_PB*	1.671	0.036	0.004	0.84
MC_ROE*	1.597	0.050	0.004	0.83
MC_TA	1.579	0.055	0.004	0.83
MC_HC	1.562	0.059	0.004	0.80
Six factor	GRS F-Statistics	P-Value	Average absolute alpha value	Average R2
MC_PB*	1.647	0.040	0.0039	0.85
MC_ROE	1.497	0.079	0.0040	0.84
MC_TA	1.537	0.066	0.0036	0.84
MC_HC	1.553	0.061	0.0039	0.81

* Significant @ 5% level